

Ministry of Municipalities and Housing

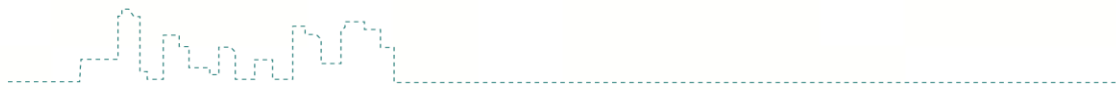
Policy AI Lab / H.E.M Office

Business Requirements Document

BRD

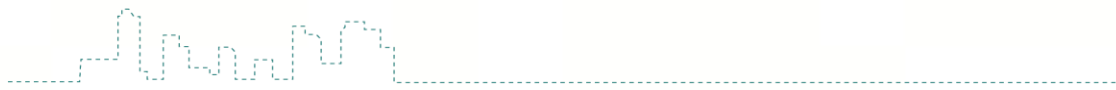
23-April-26





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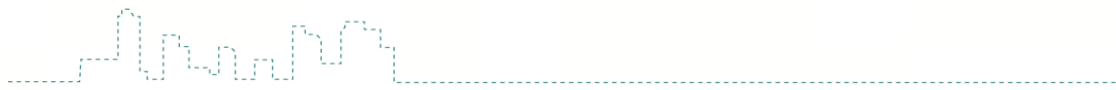


1. Document Information

Purpose: To establish a unified understanding of the new system's scope and detailed requirements, serving as the primary reference for design, development, testing, and operations.

1.1 Version History

Version	Date	Description	Author
v1.0	Feb 9 2026	Started Document	Moazzam Ahmed
v1.1	Feb 17 2026	Added Problem Statement & Strategic Opportunity	Moazzam Ahmed
v1.2	Feb 24 2026	Added In-Scope, Out of Scope	Moazzam Ahmed
v1.3	Mar 10 2026	Added Definitions	Moazzam Ahmed
V1.4	Mar 14 2026	Added Requirements	Moazzam Ahmed
V2.0	Apr 23 2026	Converted to NHC BRD format	Moazzam Ahmed



2. Executive Summary

The Ministry of Municipalities and Housing (MOMAH) is at the epicenter of Saudi Arabia's national transformation. As the Kingdom accelerates reform under Vision 2030, the complexity and interdependency of policy decisions have scaled exponentially. The Policy AI Lab is a high-performance decision-support platform designed to transition the Ministry from traditional, retrospective policymaking to an AI-augmented, predictive, and evidence-based model.

The platform solves critical challenges, including "policy silos," where initiatives in one sector (e.g., finance) inadvertently hinder another (e.g., urban development). By integrating advanced simulation and generative AI, the platform ensures that every housing policy is mathematically aligned with national KPIs, fiscally responsible, and stress-tested against global and local market volatilities.

3. Business Problem/ Challenge Articulated with Measurable Impact

The Ministry of Municipal and Rural Affairs and Housing (MOMAH) operates at the center of national urban development, municipal governance, and housing strategy. Its mandate directly influences economic diversification, infrastructure efficiency, private sector participation, and quality of life — all foundational pillars of Vision 2030.

As the Kingdom accelerates reform, the **complexity, interdependency, and scale of policy decisions have significantly increased**. Policies now:

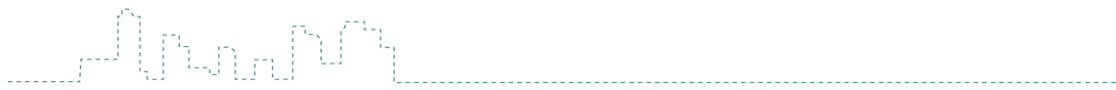
- Affect multiple sectors simultaneously real estate sector, financial sector, industrial/construction and infrastructure, Environmental and sustainability sectors (waste management, green buildings, etc.), social sectors (community services, etc.) and those related to transforming city in terms of digitization and technology adoption (smart cities, etc.).
- Carry long-term fiscal and macroeconomic implications
- Interact with global benchmarks and international competitiveness
- Require coordination across ministries and regulatory authorities
- Demand measurable, outcome-based accountability

However, the current **policy generation and review process is not structurally designed** for this level of systemic complexity.

Core Problem

MOMAH's policy lifecycle today is:

- Document-centric rather than system-centric
- Fragmented across data sources and stakeholders
- Dependent on manual analysis and siloed expertise
- Limited in predictive modeling capability



- Reactive rather than simulation-driven

3.1 Current State

MOMAH's policy lifecycle today is:

- Document-centric rather than system-centric
- Fragmented across data sources and stakeholders
- Dependent on manual analysis and siloed expertise
- Limited in predictive modeling capability
- Reactive rather than simulation-driven

3.2 Pain Points

This creates 5 Pain Points:

1. **Limited System-Wide Visibility**
Policies are developed and reviewed without a unified, canonical registry.
2. **Insufficient Predictive & Simulation Capability**
Impact assessment are sort of static reports, weak scenario analysis (particularly how risks and unintended consequences are assessed & identified), fiscal and social ripple effects are mostly either missed or not compressively analyzed to support better decision-making.
3. **Weak Institutional Memory & Feedback Loops**
Lessons from the past policies are not systematically captured, No structured mechanism to trace assumptions and outcomes, to be able to compare and learn over time.
4. **Limited Global Benchmarking Integration**
International standards, Benchmarking are not embedded into an integrated into the design and decision engine in a structured way.
5. **Extended Policy Development Timelines**
Requiring many iterations, time and bandwidth from all stakeholders.

3.3 Opportunity Statement

The proposed Policy AI Lab establishes a centralized, AI-powered decision intelligence platform that transforms MOMAH's policy lifecycle from fragmented and reactive to a unified, predictive, and evidence based.

Core opportunities include:

- **Unified Policy Registry & Knowledge Base:** Create a canonical, version-controlled registry accessible across ministries, embedding structured metadata, legal references, and sectoral linkages.



- **Predictive Modeling & Simulation:** Enable multi-scenario forecasting of fiscal, socioeconomic, and environmental impacts before enactment, reducing reversals and fiscal overruns.
- **Conflict & Duplication Detection:** Automatically flag overlapping or contradictory policies across sectors, ensuring coherence and reducing regulatory risk.
- **Benchmarking & Global Competitiveness:** Integrate international standards, peer-country comparisons, and competitiveness indices into policy design, aligning with Vision 2030.
- **Institutional Memory & Learning:** Build structured data lineage and feedback loops, allowing MOMAH to trace assumptions, measure outcomes, and continuously improve.
- **Stakeholder Engagement Intelligence:** Ingest and synthesize consultation feedback, clustering recurring themes and proposing structured amendment options.

Business Value Delivered:

- **Risk Reduction:** Institutionalized risk scoring and early detection of fiscal, legal, and operational risks.
- **Efficiency Gains:** Shortened policy development cycles (target reduction from 6–12 months to <6 months), with better quality.
- **Quality Improvement:** Fewer post-approval revisions and contradictions across ministries.
- **Vision 2030 Alignment:** Policies systematically scored against national housing and municipal KPIs, ensuring measurable contribution to reform goals.

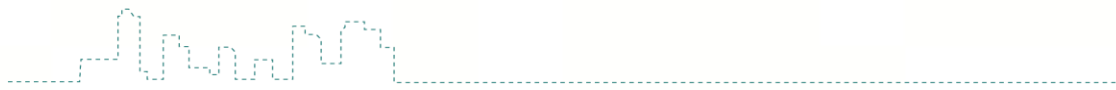
The Policy AI Lab is not just a technical system; it is a strategic enabler of faster, smarter, and more accountable policy-making, embedding simulation, benchmarking, and institutional learning into MOMAH’s core governance processes.

3.4 Strategic Alignment

<i>Vision 2030 Objective</i>	<i>The Policy AI Lab directly supports Vision 2030’s objective to enhance government effectiveness through data-driven policymaking, improved regulatory coherence, and accelerated reform delivery. By embedding simulation, benchmarking, and predictive analytics, the platform strengthens MOMAH’s contribution to national priorities such as increasing homeownership to 70%, improving Quality of Life, and ensuring fiscal sustainability through evidence-based policy design.</i>
<i>MOMAH Sector Strategic Objectives (SSOs)</i>	<i>The solution advances MOMAH’s strategic objectives by strengthening evidence-based policy development, improving regulatory quality, and reducing policy contradictions and sectoral fragmentation. It accelerates policy development cycles, enhances housing affordability and supply planning through advanced simulation, and builds institutional memory to support long-term governance excellence.</i>
<i>MOMAH AI Strategy Alignment</i>	<i>The Policy AI Lab is fully aligned with MOMAH’s AI Strategy by deploying AI-driven decision-support tools that enhance policy quality, predictive capability, and institutional efficiency. It operationalizes AI for governance, insight, and institutional learning through automated drafting, conflict detection, multi-variable simulation, risk scoring, and continuous model improvement — all under a human-in-the-loop governance framework.</i>

3.5 Expected Business Outcomes

The Policy AI Lab will be considered successful when it demonstrably improves policy quality, reduces risk, and accelerates reform cycles. Success will be measured through leading indicators (system adoption and operational health) and lagging indicators (real-world policy outcomes).



3.6 Current Solution Landscape

There are no automated processes for Policy launch. There is not even a system to support policy launch, hence no simulations are done.

The only way MOMAH tests a policy is by launching it into a small area of KSA.

4 Strategic and Operational Metrics

Strategic metrics measure the solution's impact on adoption, user satisfaction, and stakeholder confidence. These are typically reported to leadership and steering committees

Metric Type	Metric	Baseline	Target (6M / Y1 / Y2)	Measurement Method	Frequency	KPI Owner
User Adoption	Number of Users using the Policy platform	0	10 / 20 / 50	Logins	Daily	
Adoption	Number of policies generated and simulated through the tool	0	3 / 5 / 10 (80%)	Simulator Review Screen	Quarterly	

Business metrics quantify the operational and financial impact of the solution. Focus on process efficiency, cost reduction, quality improvements, and throughput gains

Metric Type	Metric	Baseline	Target (6M / Y1 / Y2)	Measurement Method	Frequency	KPI Owner
Avg. Time	Average time taken to move a policy from initial concept to approved draft using the AI platform.	0	10% / 30% / 50%	Timestamps at each stage of Policy Drafting	Quarterly	Dr. Darwaish



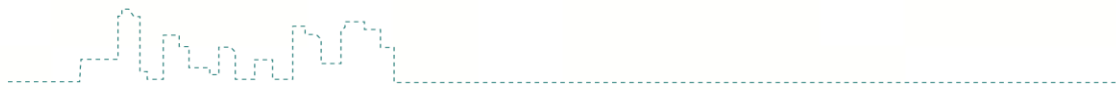
Accuracy	Policy Simulation Accuracy	0	80%	HILT Approvals without Edits	Quarterly	Dr. Darwaish
Quality	Conflict detection before drafting X% of detected conflicts flagged prior to ministerial review.	0	≥85	HILT Approvals	Quarterly	Dr. Darwaish
Quality	Assumption documentation completeness.	0	90%	Most if not all policies include structured assumption logged in the registry.	Quarterly	Dr. Darwaish
Accuracy	Risk threshold alerts: Automated alerts generated for more than X% of policies exceeding fiscal/legal thresholds.	0	>90%	# of Alerts per Policy	Quarterly	Dr. Darwaish

Lagging Indicators (Real Impact) – Happens over time

- Reduction in policy reversals
- Reduction in fiscal overruns; in budget deviations linked to policy implementation.
- Improved KPI attainment, within the targeted time-frame
- Faster reform cycles

Business Value Dimensions:

- Policy Quality Improvement: Fewer contradictions, higher evidence standards, improved independent review scores.
- Risk Reduction: Early identification of unintended consequences, fewer regulatory escalations post-launch.
- Policy Effectiveness: Higher rate of KPI achievement, faster measurable impact, improved implementation success.



Measurement & Accountability:

- Metrics will be tracked via system analytics (simulation logs, conflict detection reports, risk scoring dashboards).
- Fiscal and KPI outcomes will be validated against Ministry of Finance data and Vision 2030 dashboards.
- There should be an independent audit to assess policy quality and benchmarking accuracy annually.

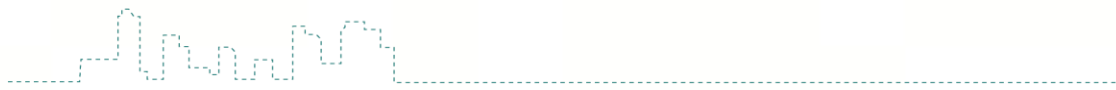
5. Vision and Objectives

Vision: To transform MOMAH's policy lifecycle from a fragmented, document-centric, and reactive process into a unified, AI-augmented, predictive, and evidence-based decision-intelligence capability — ensuring every housing and municipal policy is measurably aligned with Vision 2030, fiscally responsible, and stress-tested against global and local volatilities before enactment.

Objectives:

- Establish a canonical, version-controlled Policy Registry that serves as the single source of truth for all MOMAH policies, with full metadata, legal references, and sectoral linkages.
- Reduce the average policy development cycle from 6–12 months to under 6 months by automating drafting, conflict detection, and impact analysis.
- Embed predictive and simulation-driven decision-making so every policy is tested for fiscal, social, and market impact across multiple scenarios prior to enactment.
- Institutionalize Vision 2030 alignment by scoring each policy against national KPIs — in particular the 70% homeownership target, Quality of Life, and fiscal sustainability.
- Eliminate policy silos through cross-sector conflict detection and ripple-effect mapping across housing, municipal, fiscal, environmental, and smart-city domains.
- Build an Institutional Memory Vault that captures every policy iteration, assumption, decision, and outcome — enabling continuous learning and auditable governance.
- Enforce human-in-the-loop governance with tiered escalation (Analyst → Director → Minister) so no AI recommendation reaches enactment without expert validation.
- Achieve full compliance with Saudi PDPL, NCA Essential Cybersecurity Controls, and SDAIA AI Ethics Guidelines, with all AI inference and data storage remaining inside the Kingdom.
- Deliver role-based tools for every user persona — from the Minister's Executive Intelligence Cockpit to the Policy Analyst's drafting sandbox — grounded in explainable AI (XAI).
- Operationalize benchmarking against peer countries (e.g., Singapore, Norway) and global standards so MOMAH policies remain internationally competitive.

6. Scope



6.1 In Scope

The Policy AI Lab will deliver capabilities in three phased modules, each building on the previous. The system will include:

Policy Registry & Drafting

- Integrated & Unified Policy Registry & Versioning: Canonical repository with metadata (title, objectives, sectors, legal basis, KPIs, effective date).
- Policy Project Concept Builder including

○ Validation of alignment with sector strategies and higher-level national strategies.

- Automatically generate policy memoranda including Legal basis, Justification and objectives, Key provisions, Financial, economic, and social impact summaries, and risk considerations
- Propose recommended policy draft
- Amendments to existing policy with structured justification and traceability to source changes.
- Institutional Memory: Storing policy iterations with full explainability of why specific levers were chosen.

Benchmarking & Conflict Detection

● Policy Benchmarking & Comparative Analysis

- Benchmark policies against peer countries and global standards, contextualized for Saudi economy.
- Suggest amendments and highlight gaps and risks.

● Policy Conflict & Duplication Detection

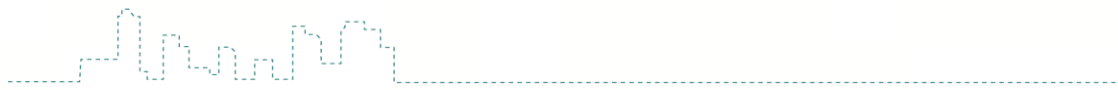
- Identify overlapping, redundant, or contradictory policies.
- Flag potential regulatory conflicts across sectors.

○ Provide impact mapping to affected policies.

● Cross-Sector Ripple Effect Mapping:

- Highlight interdependencies across housing, municipal, fiscal, and other economic sectors.

Policy Impact Simulation



- Testing "what-if" scenarios across economic, social, and municipal variables.
- Simulate multiple policy scenarios.
- Identify associated risks and trade-offs.
- Provide sensitivity analysis based on configurable assumptions.

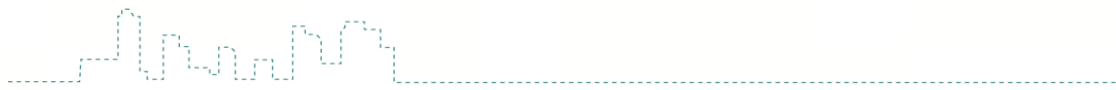
- **Consultation Comment Synthesis**

- Ingest public and stakeholder consultation feedback.
 - Cluster themes and recurring issues.
 - Summarize key concerns.
 - Propose structured amendment options based on feedback
- Risk Scoring Framework:
 - Formalized taxonomy (Legal, Fiscal, Operational, Strategic, International) with escalation thresholds.

6.2 Out of Scope

The following capabilities are explicitly excluded from the current phase:

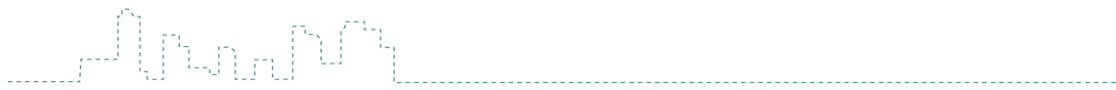
- **Autonomous Policy Approval**
 - The system will not have authority to approve, enact, or publish policies.
 - All final decisions remain human-led.
- **Binding Legal Interpretation**
 - The system does not provide legally binding interpretations or replace formal legal counsel.
 - Outputs are advisory and require expert validation.
- **Direct Legislative Draft Submission**
 - No automatic submission to legislative bodies or gazette systems without human validation and approval workflows.



- **Real-Time Macroeconomic Forecasting Infrastructure**
 - The system will not build or replace national econometric forecasting platforms.
 - It will rely on predefined models, assumptions, or integrated external datasets.
- **Full Regulatory Enforcement Monitoring**
 - The system will not monitor real-time compliance or enforce regulatory actions.
 - No inspection automation or enforcement execution.
- **Custom Model Training from Scratch**
 - The system will not develop foundational LLMs or large econometric models from scratch.
 - It will leverage existing AI models and configurable frameworks.
- **Political Decision-Making**
 - The system does not determine political priorities or make value-based policy choices.
 - Strategic direction remains with policymakers.
- **Operational Municipal Service Delivery:** The Lab will not manage day-to-day municipal operations (e.g., waste collection, utilities).
- **Citizen-Facing Applications:** The Lab is an internal decision-support tool, not a public-facing service.

7. AI Strategic Goals

#	Strategic Goal	Metric	Baseline	Target	Timeline
01	Establish AI as a Trusted Decision-Making Engine for Policy	Overrides or re-writes of policy	N/A	0	6 months
02	Accelerate Policy Innovation Through AI-Driven Experimentation	Higher Impact	N/A	TBD	6 months



03	Institutionalize Knowledge Through AI-Powered Policy Intelligence	No Siloed Data Sources	N/A	100%	
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8. Solution Overview

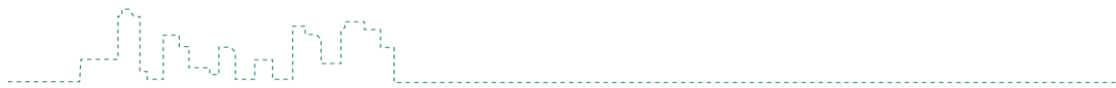
This section provides a high-level description of the proposed solution, its capabilities, target users, workflows, and operating assumptions. It should be understandable by both technical and non-technical stakeholders.

8.1 Solution Description

The Policy AI Lab is a high-performance decision-support platform designed to transition the Ministry from traditional, retrospective policymaking to an AI-augmented, predictive, and evidence-based model.

8.2 User Personas

Persona	Description
Minister	His Excellency, the CEO of MOMAH
Senior Leader / Executive	High-level decision-makers within the Executive Office .
Policy Analyst	The primary "strategist" responsible for drafting and refining policy frameworks.
Housing Economist	Expert focused on the financial and macroeconomic impact of policy levers.
Data Scientist / AI Unit	The technical team responsible for the platform's underlying models.
Legal & Compliance Officer	The gatekeeper ensuring all drafts adhere to Saudi Law and Royal Decrees.
Business Owner / Product Manager	Responsible for the strategic direction and delivery of the solution.

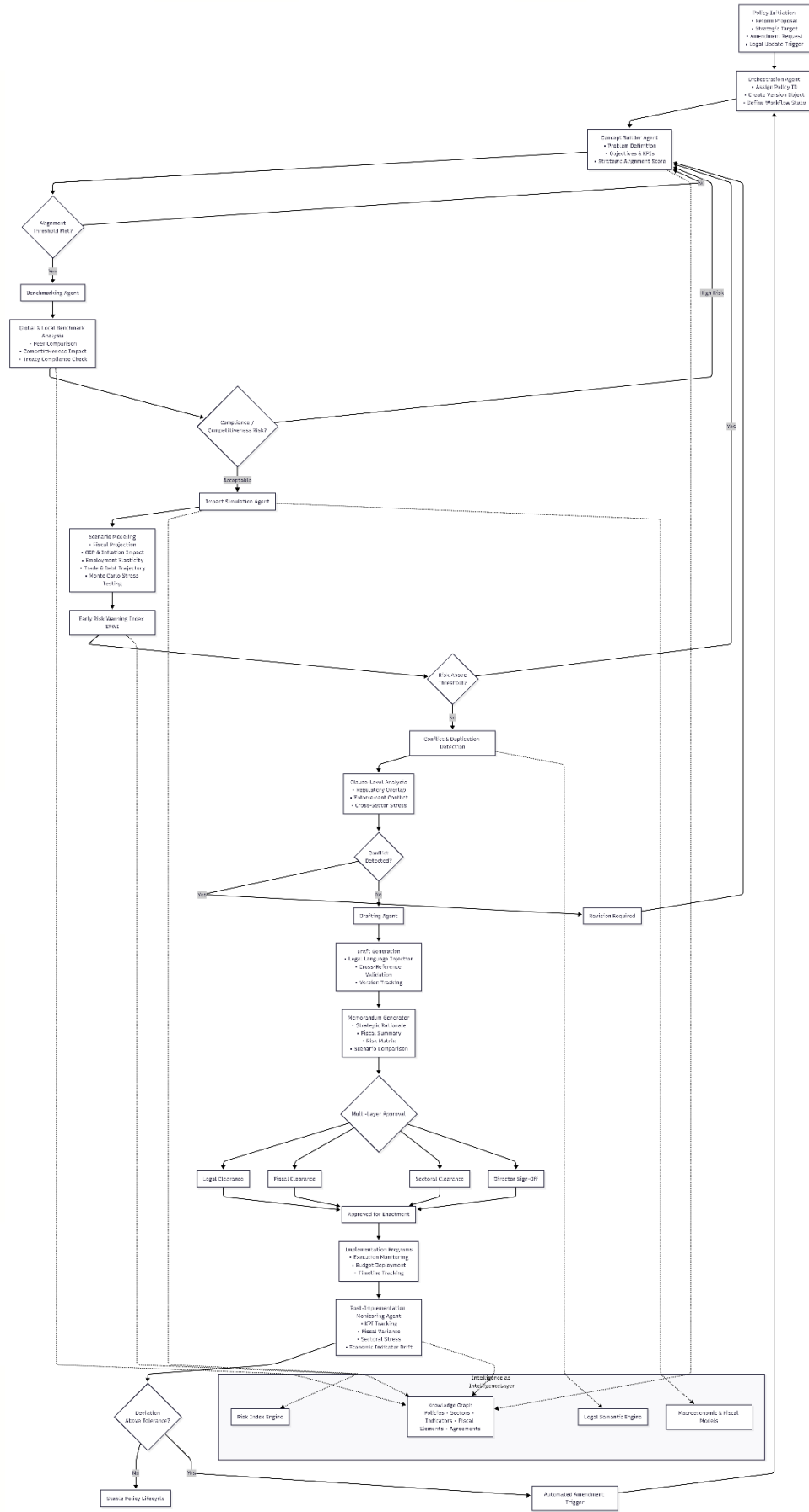


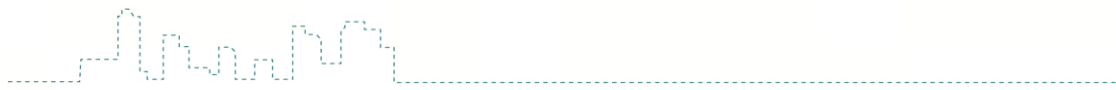
Persona	Description
Stakeholder (Internal/External)	Representative from other ministries or the public providing feedback.

8.3 Key Capabilities

ID	Capability	Functional/ Description
CAP-01	Policy Intelligence & Knowledge Layer	- Central Policy Registry & Versioning - Semantic Mapping & Knowledge Graphs - Evidence Synthesis & Research Integration
CAP-02	AI-Assisted Policy Generation	- Policy Drafting & Brief Generation - Comparative Policy Analysis - Scenario-Based Recommendations
CAP-03	Policy Simulation & Scenario Modeling	- Multi-Variable Simulation Engine - Scenario Testing - Predictive Impact Forecasting - Policy Stress Testing
CAP-04	Collaboration & Workflow Orchestration	- Multi-Stakeholder Collaboration - Policy Lifecycle Management - Auditability & Governance

8.4 Process / Workflow





8.5 User Journey

Role-Based Functionality

1. Minister (His Excellency)

Required Functionality

- Can be linked to HEM's Executive Cockpit with national-level KPIs, risk posture, and policy health indicators.
- Vision 2030 alignment scoring and scenario comparison.
- Tier-3 escalation alerts with concise decision summaries.
- Final approval gateway with override capability.

Examples:

- **Scenario Comparison Panel:**

The Minister can compare:

- *Baseline:* No policy intervention
- *Proposed Policy:* Subsidy + land release
- *Optimized Scenario:* Adjusted subsidy + developer incentive

Each scenario shows impact on homeownership, fiscal cost, and regional distribution.

- **Tier-3 Alert Example:**

"Black Swan simulation indicates a 22% drop in developer participation under current incentive structure. Immediate review recommended."

- **Executive Cockpit Example:**

A single screen showing:

- Homeownership trajectory vs. Vision 2030 target
- Top 5 high-risk policies
- Regions with highest affordability stress
- Policies requiring ministerial action this week

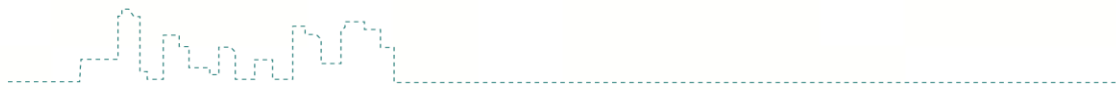
Feedback:

The Minister's interface must be **ultra-high-level**, with the ability to drill down only when needed. Recommend adding a **"Decision Justification Generator"** that produces a one-page rationale for each approval.

2. Senior Leaders / Executives

Required/Recommended Functionality

- Vision 2030 Alignment Auditor
- Regional Impact Visualization
- Human-in-the-Loop validation
- Tier-2 strategic briefings



- Cross-sector ripple effect mapping

Examples:

- **Regional Impact Visualization:**

A heatmap showing how a mortgage subsidy affects:

- Riyadh: e.g. +6% homeownership
- Jeddah: e.g. +2%
- Northern Borders: No impact due to low developer presence

- **Ripple Effect Example:**

A municipal zoning change triggers:

- Increased infrastructure cost (MoF)
- Higher land values (SAMA)
- Reduced affordability (MOMAH)

- **Tier-2 Briefing Example:**

“Policy X improves homeownership by 4% but increases fiscal exposure by SAR 1.2B. Consider adjusting LTV ratios.”

Feedback

Executives need a **policy portfolio view** showing how all active policies interact, overlap, or conflict across ministries.

3. Policy Analysts (Strategists)

Required Functionality

- Intelligent Policy Generator & Synthesizer
- Gap & Opportunity Analysis
- Consultation Comment Synthesis
- Access to Institutional Memory Vault
- Drafting sandbox with version control

Examples:

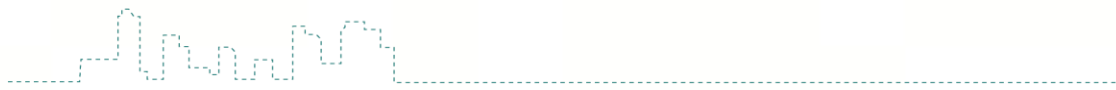
- **AI Drafting Example:**

Analyst inputs:

- Objective: “Reduce land cost by 10%”
- Constraint: “Fiscal neutrality required”

AI generates:

- Policy mechanism options (e.g., land value capture, zoning incentives)
- Legal basis referencing relevant Royal Decrees
- Expected fiscal impact
- International benchmarks (e.g., Singapore’s land acquisition model)



- **Gap Analysis Example:**
System identifies that MOMAH lacks policies addressing:
 - Developer financing bottlenecks
 - Affordable rental supply
 - Green building compliance enforcement
- **Consultation Synthesis Example:**
From 1,200 public comments, the system clusters:
 - 32% affordability concerns
 - 21% developer incentives
 - 18% zoning restrictions
 - 12% environmental impact

Feedback

Analysts need a **structured assumption builder** to ensure all policies follow a consistent logic model (e.g., population growth, material cost assumptions, SIBOR baseline).

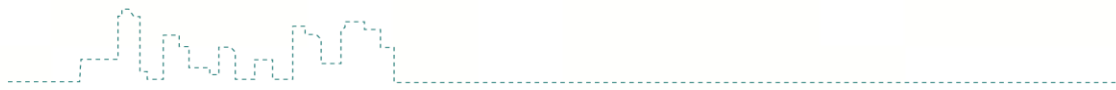
4. Housing Economists

Required Functionality

- Multi-Variable Simulation Lab (ABM)
- Black Swan stress testing
- Sensitivity analysis
- Risk escalation monitoring
- Ability to manipulate macro, social, and market variables

Examples:

- **ABM Example:**
The system simulates:
 - 1.2M households
 - 4,000 developers
 - 30 banks
 - 13 regionsEach agent reacts to policy levers (LTV changes, subsidies, land tax adjustments).
- **Black Swan Example:**
 - Global supply chain disruption ... material costs +40%
 - Interest rate shock ... SIBOR +300bps
 - Developer insolvency risk increases by 18%



- **Sensitivity Analysis Example:**

Tornado chart shows the most sensitive variables:

- Interest rates (highest impact)
- Land availability
- Developer incentives

Feedback

Economists need:

- **Exportable simulation datasets** for offline econometric modeling
- **Variable locking** to isolate single-factor impacts
- **Regional overrides** (e.g., different land values in Tabuk vs. Riyadh)

5. Data Scientists / AI Unit

Required Functionality

- Model retraining
- Explainability audits
- Drift detection
- Access to raw logs
- Threshold and guardrail configuration
- Model versioning and lineage tracking

Examples:

- **Drift Detection Example:**

“Model accuracy dropped from 92% to 83% due to new mortgage behavior among 25–34 age group.”

- **Explainability Example:**

Clicking on a simulation result reveals:

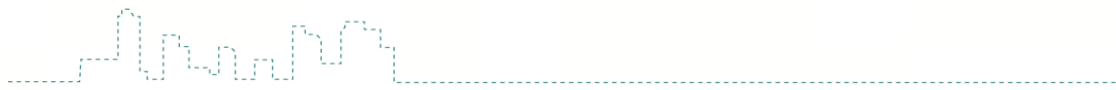
- Top 5 variables influencing outcome
- Data sources used
- Weighting logic
- Historical back-testing accuracy

- **Model Governance Example:**

Data scientists can compare:

- Model v1.2 (pre-COVID behavior)
- Model v2.0 (post-COVID behavior)
- Model v3.1 (after new SAMA regulations)

Feedback



A **model governance dashboard** is essential to ensure transparency, auditability, and compliance.

6. Legal & Compliance Officers

Required Functionality

- Regulatory Conflict Engine
- Legal basis extraction
- Compliance Gateway (PDPL/NCA)
- Clause-level traceability
- Override permissions

Examples

- **Conflict Detection Example:**
"Clause 4.2 contradicts Royal Decree 123/2019 regarding land taxation."
System highlights the exact conflicting text.
- **Legal Basis Extraction Example:**
AI identifies relevant articles from:
 - Municipal Law
 - Housing Regulations
 - Royal Orders
 - SAMA/Central bank guidelines
- **Compliance Gateway Example:**
System blocks a draft that includes personal data without PDPL justification.

Feedback

Recommend adding:

- **Legal precedent explorer**
- **Risk classification layer** (low/medium/high legal exposure)

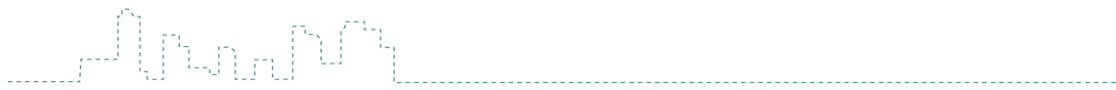
7. Business Owner / Product Manager

Required Functionality

- Policy Lifecycle Orchestrator
- Workflow management
- System usage analytics
- KPI and threshold configuration
- Role-based access control

Deeper Examples

- **Workflow Example:**
Draft ... Simulation .. Legal Review .. Executive Review .. Minister Approval
System shows bottlenecks (e.g., "Average delay: 7 days in Legal Review").



- **Usage Analytics Example:**

- 82% of policies used simulation
- 67% triggered conflict detection
- 12% escalated to Tier-2

Feedback

A **system health dashboard** and **A/B testing capability** would support continuous improvement.

8. Stakeholders (Internal/External)

Required Functionality

- Consultation portal
- Feedback submission
- View summarized responses
- Track how feedback influenced drafts

Examples:

- **Feedback Traceability Example:**

"Your comment on developer incentives was incorporated into Clause 3.1."

- **Stakeholder Segmentation Example:**

System categorizes feedback by:

- Developers
- Citizens
- Municipalities
- Banks
- NGOs

Feedback

A **transparency mode** will strengthen trust and improve consultation quality.

9. System / Platform Role

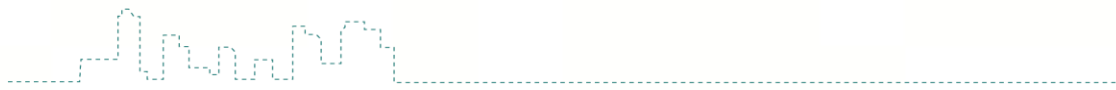
Required Functionality

- Schema enforcement
- Version control
- Knowledge graph management
- Security & access control
- Audit trail logging

Examples:

- **Schema Enforcement Example:**

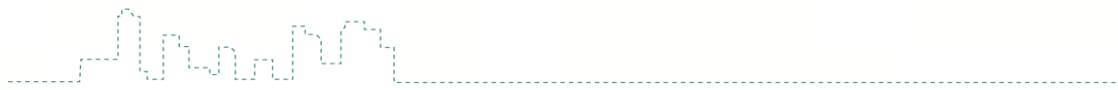
System blocks a draft missing:



- KPIs
- Issuing authority
- Policy mechanisms
- Vision 2030 mapping
- **Knowledge Graph Example:**
Policy X links to:
 - 3 Vision 2030 KPIs
 - 2 Royal Decrees
 - 4 related policies
 - 2 external ministries

Feedback

Automated schema validation and **data freshness checks** should be mandatory before a policy progresses.



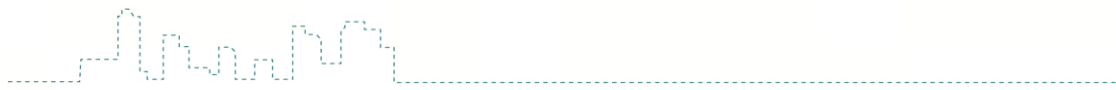
9. Stakeholders and Roles

9.1 Key Roles

Level of Impact	Role and Responsibility	Person	Email
High - Business Owner	The initiative owner and operational decision-maker approves the scope and requirements.	Dr. Ahmed Darwaish – Policy Department	
High - Executive Sponsor	Strategic oversight and approval of final outputs	Licensing and Project Coordination Agency	
High - level AI governance	Adopting models and monitoring the quality of outputs and ensuring compliance	Central AI Team	
Medium - Data Security	Ensuring data security, access standards, and protection of personal data	Cybersecurity Management	
Beneficiary - Primary User	Applying checklists in field monitoring and licensing	Municipalities and Secretariats in the regions	

10. Terms and Definitions

Term	Definition
Policy Instrument	The specific mechanism (tax, subsidy, regulation) used to achieve an objective.
Fiscal Neutrality	A state where policy implementation costs are fully offset by generated economic value or savings.
Policy Knowledge Graph	A structured digital map of the relationships between different regulations, entities, and outcomes.



11. Integrations and Data Sources

11.1 Key Integrations

The Policy AI Lab will integrate with the following internal and national platforms to enable secure authentication, inter-ministerial data exchange, sovereign AI inference, and notification delivery. Data-only sources (GASTAT, SAMA, OECD, IMF, World Bank, etc.) are catalogued separately in §11.2 Data Sources.

- 1- **Nafath — National Single Sign-On:** Citizen and official authentication, aligned with NFR-01 (Multi-Factor Authentication) and NCA access-control requirements.
- 2- **SDAIA Sovereign AI Platform:** Hosting environment for all AI inference, model training, and Knowledge Graph operations within the Kingdom, per NFR-06 (Sovereign Compliance Layer).
- 3- **Yesser Government Service Bus (GSB):** Secure inter-ministerial data exchange with SAMA, Ministry of Finance, MISA, and other regulatory authorities.
- 4- **Balady Municipal Platform:** Municipal permits, zoning data, and urban planning records used for policy impact simulation at the Amana level.
- 5- **Sakani Housing Platform:** Housing application, ownership, and subsidy data feeding the Housing Economist's Multi-Variable Simulation Lab.
- 6- **Absher Public Services Gateway:** Channel for public consultation feedback ingestion into the Consultation Comment Synthesis module.
- 7- **MOMAH Enterprise Directory (AD / IAM):** Internal identity, role, and permission management mapped to the RACI matrix in the Appendix.
- 8- **Document Management System (DMS):** Immutable storage for generated Policy Memoranda, Impact Briefs, Decision Decks, and the full audit trail.

11.2 Data Sources

Data Source	Data Type	Source system	Owner	Sensitivity	Notes
Legal Repository	New Internal Policy Data Source	New	Dr. Ahmed Darwaish	Confidential / Internal	
HDOC (under BIDSC)	Housing demand & occupancy data	Internal — MOMAH / BIDSC	BIDSC	Confidential / Internal	Primary feed for baseline supply-demand and affordability modelling.
REGA Database	Real-estate regulatory & transaction data	Ecosystem (outside Ministry)	Real Estate General Authority	Restricted	Accessed via inter-entity data-sharing agreement.
GASTAT	Demographic, census & labour statistics	External — Government	General Authority for Statistics	Public / Open	Primary source for household formation, urbanization, and demographic layers of the Simulation Engine.



SAMA (Saudi Central Bank)	Mortgage, SIBOR, and monetary-policy indicators	External — Government	Saudi Central Bank	Restricted	Triggers risk thresholds under the Market Volatility and Affordability Drift categories.
Ministry of Finance	Macroeconomic forecasts, fiscal envelopes, NPV baselines	External — Government	Ministry of Finance	Confidential	Required for Fiscal Variance monitoring and Strategic Waiver workflows.
Vision 2030 PMO	National KPI definitions & trendlines	External — Government	Vision 2030 Realisation Office	Official Use	Baseline for the Vision 2030 Alignment Auditor (CAP-02).
OECD Data	International benchmarking & policy indicators	External — Global	OECD	Public / Open	Used by Policy Benchmarking & Comparative Analysis module.
IMF Data	Macroeconomic forecasts & country comparisons	External — Global	International Monetary Fund	Public / Open	Cross-referenced for Black-Swan and stress-test scenarios.
World Bank Open Data	Global housing, governance, and urban indicators	External — Global	World Bank	Public / Open	Benchmarking against peer countries (Singapore, Norway, etc.).

12. Business Requirements

12.1 Functional Requirements

ID	Feature Name	Business Requirement Description	User Persona	Priority
01	Policy Entity	A policy object to be utilized as the base object for the system. <i>As defined by Artifact 1.0</i> UAC: GIVEN a new policy is created WHEN saved THEN it must conform	System	High



ID	Feature Name	Business Requirement Description	User Persona	Priority
		to the defined Policy Object schema (ID, sector, KPIs, metadata) AND be stored in the registry		
02	Intelligent Policy Generator & Synthesizer	AI-assisted drafting of policy frameworks grounded in Saudi Law, Royal Decrees, and global best practices. <i>Based on Policy Entity and Definitions in Artifact 2.0</i> UAC: GIVEN a user inputs objectives and constraints WHEN generation is triggered THEN the system produces a structured policy draft aligned with legal and national frameworks AND includes justification, impacts, and provisions	Policy Analyst	High
03	Regulatory Conflict Engine	Automated scanning of new drafts to detect duplication, overlaps, contradictions, or legal friction with existing national and municipal regulations. <i>As defined in Artifact 2.0.</i> UAC: GIVEN a policy draft WHEN analyzed THEN the system flags conflicts, overlaps, or contradictions AND provides references to conflicting policies	Legal Officer	High



ID	Feature Name	Business Requirement Description	User Persona	Priority
04	Policy Gap & Opportunity Analysis	AI-driven identification of regulatory "white spaces" where new policy interventions are required to meet Vision 2030 targets. UAC: GIVEN existing policy landscape WHEN analyzed THEN the system identifies gaps AND suggests areas for new policy interventions	Strategist	High
05	Vision 2030 Alignment Auditor	Real-time scoring of policy drafts against National Transformation Program (NTP) KPIs and homeownership targets. Need to define or link these below UAC: GIVEN a policy draft WHEN evaluated THEN the system assigns a quantitative alignment score AND shows contribution to Vision 2030 KPIs	Senior Leader	High
06	Institutional Memory Vault	A searchable repository of all historical policy iterations, decisions, and precedents with AI-powered "explainability" for every recommendation. UAC: GIVEN a user searches historical policies WHEN accessed THEN the system displays previous versions, decisions, and AI explainability for recommendations	Analyst / Audit	High



ID	Feature Name	Business Requirement Description	User Persona	Priority
07	Regional Impact Visualization	Geospatial heatmaps showing how national policies manifest differently across various Saudi Amanas and regions. UAC: GIVEN a policy simulation WHEN viewed THEN the system displays regional impact via h	Senior Leader	Medium
08	Web based platform	Role based portal for all user personas to have access to appropriate functionality UAC: GIVEN a user logs in WHEN authenticated THEN they can access features based on role permissions securely	Everyone	High
09	Policy Simulation	Capability to run "Black Swan" simulations (e.g., global material spikes, interest rate shocks) to test policy resilience. <i>As defined by Artifact 3.0</i> UAC: GIVEN a policy scenario WHEN simulated THEN the system models extreme conditions AND outputs risk and impact analysis	Economist	Medium
10	Multi-Variable Simulation Lab	Agent-Based Modeling (ABM) to simulate policy impact across economic (SIBOR, GDP), social	Economist	High



ID	Feature Name	Business Requirement Description	User Persona	Priority
		(Demographics), and market (Supply/Demand) variables. UAC: GIVEN configurable variables WHEN simulation runs THEN the system models interactions across economic, social, and market variables AND produces measurable outputs		
11	Predictive Success Scoring	Probabilistic forecasting of a policy's likelihood to achieve its stated objectives based on historical and simulated data. AI must fail the Simulation if at least 80% success rate is not achieved, citing reasons for failure. <i>As defined in Artifact 4.0</i> UAC: GIVEN a simulation result WHEN evaluated THEN the system assigns a probability score AND flags failure if success <80% with explanation	Data Scientist	Medium
12	Risk Escalation	AI must predict risk and escalate a policy that jeopardizes Policy success criteria. <i>As defined in Artifact 5.0</i> UAC: GIVEN a policy exceeds risk thresholds WHEN detected THEN the	Economist	High



ID	Feature Name	Business Requirement Description	User Persona	Priority
		system triggers escalation workflows and alerts based on severity tiers		
13	Policy Lifecycle Orchestrator	Management of the end-to-end workflow from initial AI-drafting to simulation, expert review, and final executive approval. UAC: GIVEN a policy lifecycle WHEN initiated THEN the system enforces stages (draft → simulation → review → approval) with tracking	Product Manager	High
14	Human-in-the-Loop Gateway	A mandatory governance interface for expert validation, manual overrides, and "kill-switch" triggers before policy finalization. UAC: GIVEN a policy reaches validation stage WHEN reviewed THEN experts can approve, reject, or override AI outputs AND all actions are logged	Senior Leader	High
15	Continuous Learning Loop	Automated retraining of simulation models based on real-world outcome data to improve future predictive accuracy. UAC: GIVEN real-world policy outcome data WHEN ingested THEN the system updates models AND improves prediction accuracy over time	Data Scientist	Medium



12.2 Non-Functional Requirements (NFRs)

NFR-01	Security	Multi-factor authentication via Nafath; end-to-end encryption.
NFR-02	Performance	Simulations must provide "Level 1" results in < 60 seconds.
NFR-03	Scalability	Ability to process billions of data points as the Knowledge Graph grows.
NFR-04	Language	Arabic and English
NFR-05	Explainable AI (XAI) Framework	Every AI-generated clause or simulation outcome must provide a "reasoning path," citing the specific data points, laws, or historical precedents used.
NFR-06	Sovereign Compliance Layer	All AI inference and data storage must be conducted within the Kingdom, ensuring 100% adherence to NCA and SDAIA AI ethics and residency guidelines.



1. Policy Knowledge Framework (Artifact 1.0)

The platform represents policies as a multi-dimensional "Knowledge Graph." Each policy entry must be tagged with the following metadata to enable precise simulation and alignment checks:

a. Core Policy Metadata (The "Policy Object")

Every policy stored in the system is defined by a mandatory schema that ensures interoperability across Ministry departments:

1. **Policy Identification:**
 - i. **Policy Name & ID:** Unique identifier for version control and cross-referencing.
 - ii. **Issuing Authority:** The specific department or entity responsible for the mandate.
 - iii. **Effective Date & Expiry:** Temporal bounds to track policy relevance.
2. **Strategic Categorization:**
 - i. **Policy Sector:** Primary domain (e.g., Housing, Municipal, Infrastructure).
 - ii. **Sub-Sector:** Granular classification (e.g., Waste Management, Mortgage Finance).
 - iii. **Vision 2030 Pillar:** Direct mapping to national goals (e.g., Quality of Life, Housing 70%).
3. **Functional Components:**
 - i. **Objectives:** The primary "North Star" (e.g., "Decrease land cost by 10%").
 - ii. **Policy Mechanisms:** The specific levers used (e.g., Subsidies, Taxation, Land Grants).
 - iii. **Implementation Constraints:** Budgetary limits, geographic restrictions, or legal boundaries.

b. Semantic Mapping & Relational Entities

In addition to **Stakeholders** and **Regulatory Dependencies**, every "Policy Object" must be mapped to the following relational entities:

1. **Strategic Alignment Anchors (Vision 2030 Mapping):**
 - i. **Directives:** Links to specific National Transformation Program (NTP) mandates.
 - ii. **KPI Linkages:** Direct relationship to national targets (e.g., mapping a mortgage policy directly to the *70% Homeownership Rate*).
 - iii. **Pillar Association:** Vertical mapping to pillars such as *Quality of Life* or *Economic Diversification*.
2. **Cross-Sectoral Interdependencies:**



- i. **Horizontal Links:** Relationships with other sectors (e.g., linking a "Housing" policy to "Energy" via green building standards or "Transport" via urban density mandates).
- ii. **Inter-Ministerial Nodes:** Formal connections to external authorities like **SAMA** (Financial), **MoF** (Fiscal), or **MISA** (Investment).
- iii. **Geospatial & Regional Entities:**
- iv. **Impact Zones:** Mapping policies to specific **Amanas** (Municipalities) or regions where the policy is active.
- v. **Urban Context:** Distinguishing between urban "Metro" centers (Riyadh/Jeddah) and rural or industrial zones.

3. Economic & Market Anchors:

- i. **Variable Sensitivity:** Linking the policy to specific external indices (e.g., SIBOR, Construction Material Index, CPI) that act as "Triggers" for policy failure or success.
- ii. **Market Segments:** Mapping to specific sub-sectors like *Affordable Housing*, *Luxury Real Estate*, or *Commercial Municipal Zoning*.

4. Fiscal & Funding Sources:

- i. **Budgetary Origin:** Identifying if the policy is funded via the **National Budget**, **PIF**, or private-sector participation (PPP).
 1. **Revenue Nodes:** Identifying if the policy generates revenue (e.g., White Land Tax) and where those funds are re-allocated.
- ii. **Operational Triggers (Policy Logic):**
- iii. **Conditions:** The "If-Then" logic (e.g., *If interest rates rise > 4%, then the subsidy mechanism increases by 0.5%*).
- iv. **Institutional Memory Tags:** Links to previous historical policies that failed or succeeded under similar market conditions.

c. Policy Lifecycle & Audit Metadata

To support the "Institutional Memory" requirements, the framework captures the evolution of the policy:

1. **Version History:** Tracking the "DNA" of the policy through various simulation iterations.
2. **Success Metrics (Actual vs. Simulated):** Tagging policies with historical performance data to improve future AI recommendations.
3. **Human-in-the-Loop Validation:** Metadata regarding which expert analysts reviewed and approved specific clauses.



2. Policy Generation Engine (Artifact 2.0)

By utilizing this metadata, the **Policy Generation Engine** can perform the following automated functions:

- **Regulatory Conflict Detection:** The AI cross-references the **Issuing Authority** and **Regulatory Dependencies** to flag if a new draft contradicts an existing decree from a different sub-sector.
- **Automated Gap Analysis:** The system identifies "Under-Served" **Sectors** or **Sub-Sectors** where current policy objectives do not align with the desired **Vision 2030** trajectory.
- **Contextual Drafting:** When drafting, the AI pulls from the **Policy Mechanisms** of top-performing international benchmarks (e.g., Singapore or UAE) and adapts them to the **Saudi Implementation Constraints**.

3. Policy Scenario Simulation (Artifact 3.0)

A. The Input Variable Matrix

The platform must allow users to provide and manipulate variables across four distinct layers:

- **Macro-Economic Layer:**
 - **Interest Rates:** Ability to simulate fluctuations in borrowing costs for both developers and mortgage-seeking citizens.
 - **Construction Material Index:** Real-time and forecasted costs for steel, cement, and labor.
 - **GDP & Inflation (CPI):** Macro-economic stability indicators that dictate citizen purchasing power.
- **Demographic & Social Layer:**
 - **Household Formation Velocity:** The rate at which the youth population creates new independent households.
 - **Urbanization Shift:** Percentage of the population migrating toward primary Amanas (Riyadh, Jeddah, Dammam).
- **Supply & Implementation Layer:**
 - **Land "White Tax" Elasticity:** How different taxation rates on undeveloped land trigger new construction starts.
 - **Developer Incentives:** Subsidies, fast-track permit lanes, or land-grant values.
- **Financial & Market Layer:**
 - **Loan-to-Value (LTV) Ratios:** Direct control over mortgage lending stringency.
 - **Rental Yield Variance:** Simulating the shift between the rental market and ownership demand.

B. Stress Testing Methodology: The "Black Swan" Framework



Stress testing **Policy Resilience** against extreme, non-linear events.

- **Sensitivity Analysis (Tornado Testing):** The system automatically varies one input (e.g., a 200bps interest rate hike) while keeping others constant to identify which variable has the most volatile impact on homeownership.
- **Scenario-Based "Perfect Storms":** Combinations of adverse events are modeled simultaneously—for example, a simultaneous global supply chain disruption (material costs up 40%) paired with a local economic slowdown (GDP growth stalls).
- **Regional Stress Testing:** The simulation must allow for "Geographic Fencing." A policy that works for Riyadh might fail in the Northern Border region due to differing land values or developer concentrations.
- **Stochastic Modeling (Monte Carlo):** Running the simulation 10,000+ times with random variations in market behavior to identify the "Probability of Policy Failure."

C. The Assessment Protocol (The Policy Audit)

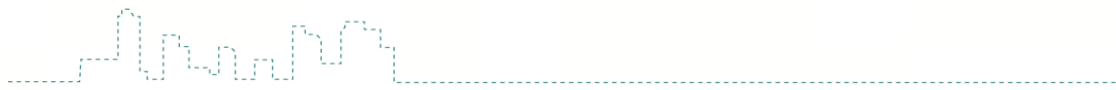
The "Assessment Layer" evaluates the output against three primary benchmarks:

- **Vision 2030 Alignment Score:** A quantitative score indicating how much closer the policy moves the Kingdom to the **70% homeownership target** and other National Transformation Program (NTP) KPIs.
- **Fiscal Impact Analysis:** A calculation of the Net Present Value (NPV) to the government. Does the policy generate long-term economic activity (GDP contribution) that outweighs the immediate cost of incentives?
- **Social Equity Impact:** An assessment of how the policy affects different "Audience Segments" or income deciles. Does it inadvertently favor high-income earners while pricing out the middle class?
- **Explainability Audit:** The AI must provide a "Traceability Path," explaining exactly *why* the simulation resulted in a specific outcome (e.g., "The land tax increase was offset by a rise in SIBOR, resulting in net-zero supply growth").

D. Defining "Simulation Success"

A simulation is deemed **successful** when it meets the following validity gates, regardless of whether the simulated policy itself is "good" or "bad":

- **Computational Convergence:** The model must reach a stable state where agents (citizens, banks, developers) reach an equilibrium without the system crashing or producing infinite loops.
- **Predictive Plausibility (Back-Testing):** When historical data from the last five years is fed into the engine, it must be able to "predict" the actual historical outcome with a high degree of



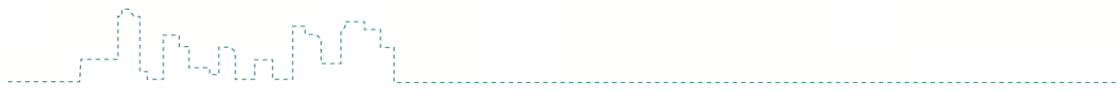
statistical accuracy (e.g., >90%).

- **Traceability & Explainability:** A simulation is only successful if it can be "interrogated." Users must be able to click on any result and see the underlying logic and data sources that led to that conclusion.
- **Actionable Delta:** The simulation must produce a measurable difference from the "Baseline/Status Quo" scenario. If the simulation cannot distinguish the impact of a new policy from doing nothing, it has failed to provide decision-support value.

4. Policy Assessment Criteria (Artifact 4.0)

Policies are scored using the following matrix:

Metric	Measurement Goal	Data Source
Vision 2030 Alignment	Policy Outcome fulfills 15% goals	Manual
National Stability	Enhance community safety & security	Manual
Economic Development	GDP growth by x%	Manual
Affordability Index	Price-to-Income ratio targeting < 5x	GASTAT / SAMA
Homeownership Rate	Delta increase toward 70%	MOMAH Records
Fiscal Impact	Net cost to the treasury over 10 years	MoF / Budgeting
Supply Velocity	Days from permit to market entry	Municipal Data



Metric	Measurement Goal	Data Source
Interdependency Mapping	Identify “Ripple Effect” – how one policy may impact other sectors	MOMAH Records

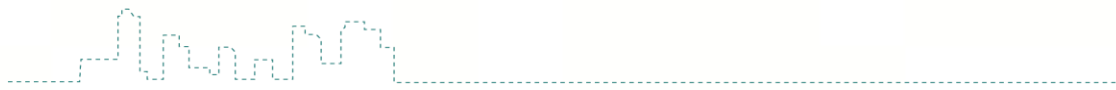
5. Risk Escalation (5.0)

To ensure the **Policy AI Lab** provides actionable governance rather than just data, the system must include a rigorous **Policy Risk & Escalation Framework**. This framework defines when a simulation indicates that a current policy is "failing" or drifting from **Vision 2030** targets, triggering a structured response.

a. Risk Escalation Thresholds (The "Red-Flag" Logic)

Thresholds are categorized by the severity of the delta between *Projected Impact* and *Target KPIs*. These are monitored via a continuous "Policy Health" scan.

Risk Category	Metric Threshold (Policy Failure Indicator)	Severity
Affordability Drift	Price-to-Income ratio exceeds 5.5x in primary regions (Riyadh/Jeddah).	Critical
Supply-Demand Gap	Projected housing supply deficit > 15% over a 24-month horizon.	High
Fiscal Variance	Policy implementation cost exceeds budget by > 10% without a 1:1 ROI.	High
Ownership Stagnation	Homeownership rate growth falls > 2% below the Vision 2030 annual trendline.	Critical
Market Volatility	Private developer participation/investment drops by > 20% year-over-year.	Medium



b. Escalation Paths (The Tiered Protocol)

When a simulation crosses a threshold, the system automatically routes the insight through an established hierarchy to ensure a "no-silent-failure" environment.

- **Tier 1: Tactical Alert (Technical/Analytical Level)**
 - **Recipient:** Policy Analysts and Data Scientists.
 - **Trigger:** Medium-severity alerts or "Yellow" status (e.g., minor supply delays).
 - **Action:** System requires the analyst to run a "Root Cause Simulation" to identify if the issue is a variable glitch or a structural policy flaw.
- **Tier 2: Strategic Briefing (Director/Chief Economist Level)**
 - **Recipient:** Director of Policy, Chief Economist, Department Heads.
 - **Trigger:** High-severity alerts or persistent "Yellow" trends.
 - **Action:** The system generates a **Policy Optimization Memo**, suggesting specific lever adjustments (e.g., increasing interest subsidies) to bring the policy back to the "Efficiency Frontier."
- **Tier 3: Executive Intervention (Minister/Cabinet Level)**
 - **Recipient:** The Minister, His Excellency's Advisors, and the Executive Office.
 - **Trigger:** Critical-severity alerts (e.g., systemic affordability collapse).
 - **Action:** Deployment to the **Minister Intelligence Cockpit** via the "Breaking News" ticker. The system prepares an emergency "Decision Deck" summarizing the failure and three simulated "Recovery Scenarios."

c. Policy Guardrails (The Operational Safety Net)

Guardrails are built-in system constraints that prevent the platform from generating or recommending "impossible" or "unstable" policy outcomes.

- **Fiscal "Hard-Stop":** The AI cannot recommend a policy mechanism where the Net Present Value (NPV) to the Treasury is negative over a 15-year term without an explicit "Strategic Waiver" from the Minister of Finance.
- **Stochastic Variance Guard:** If the simulation results show high volatility (e.g., a "Monte Carlo" failure rate >30%), the system must label the policy as "Unstable" and prohibit it from reaching the Executive Dashboard.
- **Explainability Gateway:** No policy recommendation can be escalated unless the AI can cite at least **three historical precedents** or **two legal circulars** that ground the logic. "Black-box" recommendations are automatically rejected.
- **Data Latency Guard:** If integrated datasets (e.g., SIBOR rates or GASTAT data) are older than 30 days, simulations are watermarked with a "Low Confidence" warning to prevent decisions based on stale market signals.

d. Automated Notifications & Communication



The system ensures that the right "Signal" reaches the right user through integrated delivery channels.

- **The Breaking News Ticker:** Reserved for Tier 3 executive alerts, appearing at the top of the **Executive Intelligence Cockpit**.
- **Contextual Alert Digests:** Weekly subscription-based summaries for analysts that group "policy drifts" by sector (e.g., "3 Financial Policies currently at Medium Risk").
- **Geospatial Pulse Notifications:** If a policy is failing in a specific Amana (e.g., Dammam), the regional director receives a push notification on the mobile platform with a heat map showing the localized impact.
- **Drift Detection Alerts:** For data scientists, a technical notification is sent if the "Simulation Model" is no longer reflecting real-world market outcomes, indicating a need for model retraining.

6. Data Requirements

Internal: MOMAH housing permits, Sakani application data, Balady municipal records.

External: GASTAT (demographics), SAMA (mortgage lending rates), Ministry of Finance (macroeconomic forecasts).

Global: International housing benchmarks (Singapore, Norway, etc.) for AI cross-referencing.

7. Governance and Controls

AI Explainability (XAI): The system must cite the specific data or law used to generate a recommendation.

Audit Trail: Immutable logs of every simulation run and every draft change.

Compliance Gateway: All outputs must pass an automated "PDPL" (Data Privacy) and "NCA" (Cybersecurity) check.

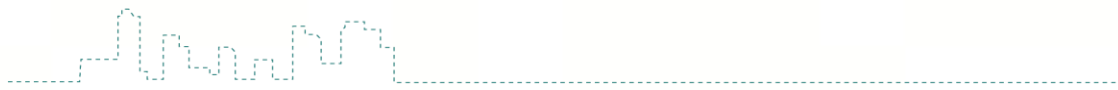
8. Reporting and Insights

Executive Cockpit: A real-time dashboard for the Minister showing the "Policy Health" of the Kingdom.

Impact Briefs: Automated 2-page PDFs summarizing simulation results for Cabinet meetings.

Anomaly Alerts: Push notifications if external market shifts (e.g., SIBOR hike) put an existing policy at risk.

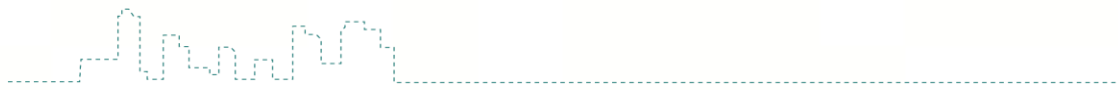
13. Agent Roles and Decision Domains (Optional)



14. Use Cases

14.1 Use Case Details

UC ID	UC-01
Use Case Name	AI-Assisted Policy Drafting
Related Functional Requirement	Refer To FR-02
Objective	Enable Policy Analysts to generate a structured, evidence-based policy draft grounded in Saudi Law, Royal Decrees, and Vision 2030 KPIs, reducing drafting time from weeks to hours.
Primary Actors	Policy Analyst
Secondary Actors	Legal & Compliance Officer; Housing Economist; AI Generation Engine (System)
Trigger / Precondition	Policy Analyst initiates a new policy in the Policy Registry with stated objectives and constraints.
Preconditions	Analyst authenticated via Nafath; Policy Registry accessible; Legal Repository and benchmark datasets synced within the last 30 days.
Main Scenario	- Analyst enters policy objective(s), target sector, and constraints (fiscal, legal, geographic). - System performs semantic search across the Knowledge Graph and retrieves relevant precedents, Royal Decrees, and global benchmarks. - AI generates a structured policy draft including legal basis, objectives, key provisions, mechanisms, and impact summary. - System attaches an Explainability Report citing the data points, legal articles, and precedents used. - Analyst reviews, edits, and saves the draft as v0.1 in the Institutional Memory Vault.
Alternative Scenarios	A1. If fewer than three precedents or two legal circulars are available, the AI flags the output as “Insufficient Grounding” and prevents escalation (Explainability Gateway). A2. If the objective conflicts with an existing ministerial directive, the system halts generation and redirects the analyst to the Conflict Detection screen. If failure occurs: the draft is quarantined, Analyst is notified, and the incident is logged in the audit trail for Data Scientist review.
Business Rules	Every generated clause must cite its supporting data source (XAI mandate). Drafts are auto-versioned and become immutable once saved. All AI inference must be performed on sovereign infrastructure (NCA / SDAIA compliance).
Data & Outputs	Inputs: Objectives, Constraints, Target Sector, Target KPIs, Budget envelope. Outputs: Structured Policy Draft (JSON + PDF), Explainability Report, Vision 2030 Alignment Score, Initial Risk Score.
Acceptance Criteria	GIVEN valid objectives and constraints WHEN “Generate” is triggered THEN a structured draft is produced within 60 seconds AND every clause includes a citation AND the Vision 2030 Alignment Score is calculated AND the draft is stored as v0.1 in the Institutional Memory Vault.



#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Policy Drafting Workspace						
Section: Policy Inputs						
1	Policy Name	Text Input	—	Mandatory	Max 200 chars; must be unique in registry	Duplicate-name inline error
2	Target Sector	Dropdown	—	Mandatory	Options: Housing, Municipal, Infrastructure, Environmental, Social, Smart Cities	—
3	Sub-Sector	Dropdown	—	Mandatory	Depends on Target Sector	—
4	Primary Objective	Text Area	—	Mandatory	Max 500 chars; plain-language objective	—
5	Vision 2030 Pillar	Multi-select	—	Mandatory	Maps to national pillars	—
6	Budget Envelope (SAR)	Number	0	Mandatory	Must be non-negative	—
7	Legal / Geographic Constraints	Text Area	—	Optional	Free-text constraints for AI grounding	—
Section: Generation Controls						
8	Generate Draft	Button	—	Mandatory	Triggers AI policy generation	Spinner + ETA
9	Grounding Mode	Radio Group	Precedent-heavy	Optional	Options: Precedent-heavy / Balanced / Novel	—
Section: Draft Output						
10	Structured Draft Preview	Text (Display)	—	—	Rendered policy draft with clauses	—



11	Explainability Report	Expandable Panel	Collapsed	—	Per-clause citations	—
12	Vision 2030 Alignment Score	Display (Gauge)	—	—	0–100, colour-coded	—
13	Save to Vault	Button	—	Mandatory	Persists as v0.1	Confirmation toast

UC ID	UC-02
Use Case Name	Policy Conflict & Duplication Detection
Related Functional Requirement	Refer To FR-03
Objective	Automatically identify overlapping, redundant, or contradictory policies across MOMAH and inter-ministerial regulations before a draft enters ministerial review.
Primary Actors	Legal & Compliance Officer
Secondary Actors	Policy Analyst; Regulatory Conflict Engine (System); Inter-Ministerial Data Feeds (SAMA, MoF, MISA)
Trigger / Precondition	A policy draft is saved or submitted for review.
Preconditions	Canonical Policy Registry is up to date; inter-ministerial linkages are resolved in the Knowledge Graph.
Main Scenario	1. System scans the draft against all active policies in the Registry. 2. Regulatory Conflict Engine flags duplications, overlaps, contradictions, and jurisdictional clashes. 3. System produces a Conflict Report with severity scoring (Low / Medium / High / Critical) and references to every conflicting article. 4. Legal Officer reviews each flag and marks it as “Resolved”, “Dismissed”, or “Requires Amendment”. 5. System locks the draft from further progression until all “Critical” flags are resolved.
Alternative Scenarios	A1. If a flagged policy is owned by another Ministry, the system generates a Cross-Sector Ripple Effect report and notifies the relevant authority via Yesser GSB. A2. If conflict resolution is contested, the flag is escalated to the Director of Policy (Tier 2). If failure occurs: the Conflict Engine logs the incident and temporarily halts ingestion of the affected registry branch.
Business Rules	No policy may advance past the “Review” stage with unresolved Critical conflicts. All conflict resolutions are logged in the immutable audit trail. Every flag must be traceable to the specific article or decree that causes the conflict.
Data & Outputs	Inputs: Draft policy; active Policy Registry; inter-ministerial ontology. Outputs: Conflict Report, Severity Score, Ripple Effect Map, Audit Log entry.
Acceptance Criteria	GIVEN a draft policy WHEN submitted for review THEN a Conflict Report is generated within 30 seconds AND no draft with Critical conflicts can proceed AND every flag is traceable to source articles.

Data Sections — UC-02

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
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Screen Title: Conflict Detection Report						
Section: Scan Header						
1	Policy Reference	Text (Display)	—	—	Draft policy under scan	—
2	Scan Timestamp	DateTime (Display)	—	—	Time of last scan	—
3	Total Flags	Display	—	—	Count by severity	—
Section: Filters						
4	Severity Filter	Dropdown	All	Optional	Low / Medium / High / Critical	—
5	Ministry Filter	Multi-select	All	Optional	Filters by owning ministry	—
Section: Flag List						
6	Flag Row	Table	—	—	Each row = one conflict with severity, source article, description	—
7	Decision	Dropdown (per row)	—	Mandatory	Resolved / Dismissed / Requires Amendment	Blocks advance if any Critical unresolved
8	Justification	Text Area (per row)	—	Conditional	Required for Dismiss on Critical	—
Section: Submission						
9	Submit Resolution	Button	—	Mandatory	Advances policy only if no unresolved Critical flags	—
10	Export Report	Button	—	Optional	PDF with flag list + decisions	—

UC ID	UC-03
Use Case Name	Multi-Variable Policy Impact Simulation



Related Functional Requirement	Refer To FR-10
Objective	Simulate the fiscal, social, and market consequences of a proposed policy across multiple scenarios before enactment, to reduce reversals and fiscal overruns.
Primary Actors	Housing Economist
Secondary Actors	Policy Analyst; Data Scientist; Simulation Engine (System); Ministry of Finance (data feed)
Trigger / Precondition	Economist selects an approved draft and launches a Simulation Run.
Preconditions	Draft has cleared Conflict Detection; integrated datasets (GASTAT, SAMA, MoF) are ≤ 30 days old; simulation variables are configured.
Main Scenario	1. Economist selects a baseline scenario and configures variables across Macro-Economic, Demographic / Social, Supply, and Financial / Market layers. 2. System executes Agent-Based Modeling across the variable matrix and runs Monte Carlo iterations. 3. System produces projected impact on homeownership rate, price-to-income ratio, fiscal exposure (NPV over 15 years), supply velocity, and regional distribution. 4. System assigns a Predictive Success Score (0–100%) and flags the policy if the score is < 80%. 5. Economist compares up to three scenarios side-by-side and exports a two-page Impact Brief PDF.
Alternative Scenarios	A1. If dataset latency exceeds 30 days, outputs are watermarked “Low Confidence” (Data Latency Guard). A2. If Monte Carlo failure rate exceeds 30%, the policy is labeled “Unstable” and blocked from the Executive Dashboard (Stochastic Variance Guard). A3. If fiscal NPV is negative over 15 years, the system requires an explicit MoF Strategic Waiver before advancing (Fiscal Hard-Stop). If failure occurs: the simulation is aborted, partial results are discarded, and the Data Scientist is alerted for model diagnostics.
Business Rules	Level-1 simulation results must return in under 60 seconds (NFR-02). All simulation runs are logged in the Institutional Memory Vault with full variable snapshots. Simulation results cannot be overridden without Human-in-the-Loop validation.
Data & Outputs	Inputs: Draft policy, variable matrix, scenario assumptions, historical baseline. Outputs: Impact Projections (fiscal / social / market), Predictive Success Score, Confidence Band, Impact Brief PDF.
Acceptance Criteria	GIVEN a policy draft and configured variables WHEN “Run Simulation” is triggered THEN Level-1 results return in ≤ 60 seconds AND the Predictive Success Score is displayed AND at least one guardrail verdict (Stable / Unstable / Low Confidence) is attached to the output.

Data Sections — UC-03

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Multi-Variable Impact Simulation						
Section: Scenario Setup						
1	Policy Reference	Lookup	—	Mandatory	Selects draft policy	—



2	Scenario Name	Text Input	Baseline	Mandatory	Max 100 chars	—
3	Variable Matrix	Composite Editor	—	Mandatory	Macro / Demographic / Supply / Financial layers	—
Section: Execution						
4	Run Simulation	Button	—	Mandatory	Launches Monte Carlo	Spinner + ETA
5	Predictive Success Score	Display (Gauge)	—	—	0–100	“Below threshold” if < 80
6	Confidence Band	Display	—	—	Upper / lower bounds	—
Section: Comparison						
7	Scenario Compare Panel	Side-by-side (up to 3)	—	Optional	Baseline / Proposed / Optimized	—
8	Export Impact Brief	Button	—	Optional	Generates 2-page PDF	—

UC ID	UC-04
Use Case Name	Risk Threshold Escalation & Tiered Alerting
Related Functional Requirement	Refer To FR-12
Objective	Ensure that a policy drifting from Vision 2030 targets triggers the correct organizational response tier (Analyst → Director → Minister) with no silent failures.
Primary Actors	System (automated Policy Health scan)
Secondary Actors	Policy Analyst (Tier 1); Director of Policy / Chief Economist (Tier 2); Minister & Executive Office (Tier 3)
Trigger / Precondition	Continuous Policy Health scan detects a threshold breach on an active or simulated policy.
Preconditions	Risk thresholds are configured; notification channels (email, mobile, Executive Cockpit ticker) are active; organizational hierarchy is mapped.
Main Scenario	1. System continuously monitors each active policy’s projected KPIs against Vision 2030 targets. 2. Upon a breach, the system classifies severity as Medium (Yellow), High (Orange), or Critical (Red). 3. Medium breach → Tier-1 alert to Analysts with a Root Cause Simulation request. 4. High breach → Tier-2 Strategic Briefing to the Director with a Policy Optimization Memo. 5. Critical breach → Tier-3 Executive Intervention to the Minister Intelligence Cockpit with a Recovery-Scenarios Decision Deck. 6. Recipients acknowledge the alert and initiate the prescribed action.



Alternative Scenarios	A1. If a Tier-1 alert is unacknowledged for 48 hours, it auto-escalates to Tier 2. A2. If a Critical alert is triggered outside business hours, a push notification is sent to the Executive Office duty officer. A3. If multiple policies breach simultaneously in the same Amana, the system aggregates them into a single Geospatial Pulse Notification. If failure occurs: the system falls back to email-only delivery and logs the channel outage for operations.
Business Rules	No Critical alert may be dismissed without a documented rationale stored in the audit trail. Tier-3 alerts appear on the “Breaking News” ticker of the Executive Cockpit. Weekly Contextual Alert Digests summarize all Yellow and Orange drifts by sector for analysts.
Data & Outputs	Inputs: Live simulation and real-world outcome data; configured thresholds; organizational hierarchy. Outputs: Tiered alert, Policy Optimization Memo, Decision Deck (Tier 3), audit log entry.
Acceptance Criteria	GIVEN a threshold breach WHEN detected THEN the correct tier recipient is notified within 5 minutes AND the alert is logged immutably AND unacknowledged alerts auto-escalate per the defined SLA.

Data Sections — UC-04

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Policy Health & Alerts						
Section: Global Health Dashboard						
1	Breaking News Ticker	Display	—	—	Tier-3 alerts only	Auto-updates
2	Policies at Risk	Tile	0	—	Count by severity	—
3	Affected Regions	Geo Tile	—	—	Regions with active alerts	—
Section: Alert Queue						
4	Alert Row	Table	—	—	Each row = one alert with severity, recipient tier, age	—
5	Acknowledge	Button (per row)	—	Mandatory	Marks alert as acknowledged	Resets auto-escalation timer
6	Escalate Manually	Button (per row)	—	Optional	Moves to next tier	Requires justification
Section: Subscriptions						



7	Weekly Digest	Toggle	On	Optional	Subscribes to Contextual Alert Digests	—
8	Geospatial Push	Toggle	Off	Optional	Mobile push for regional directors	—

UC ID	UC-05
Use Case Name	Consultation Feedback Synthesis
Related Functional Requirement	Refer To FR-02
Objective	Ingest large volumes of public and inter-ministerial consultation feedback, cluster recurring themes, and surface structured amendment options to the Policy Analyst.
Primary Actors	Policy Analyst
Secondary Actors	External Stakeholders (public / ministries); Consultation Ingestion Service (System); AI Synthesis Engine
Trigger / Precondition	A policy draft enters the “Public Consultation” stage and feedback is collected via integrated channels.
Preconditions	Consultation channel is open (Absher portal / web); minimum feedback-volume threshold reached; PDPL compliance check passed on ingestion pipeline.
Main Scenario	1. System ingests comments from Absher portal, email submissions, and inter-ministerial responses via Yesser GSB. 2. AI Synthesis Engine clusters feedback into thematic groups (e.g., affordability, developer incentives, zoning, environmental impact). 3. System produces a Consultation Report with theme weights, sentiment scoring, and representative excerpts. 4. System proposes structured amendment options per theme, each linked to the affected clause in the draft. 5. Analyst accepts, modifies, or rejects amendment proposals; accepted amendments generate a new draft version.
Alternative Scenarios	A1. If feedback contains personally identifiable information, the system auto-redacts prior to analysis (PDPL compliance). A2. If a theme represents more than 25% of total feedback, it is flagged as “Material” and auto-escalated to Senior Leaders. If failure occurs: the ingestion pipeline pauses, accumulated feedback is quarantined, and the Data Scientist is alerted.
Business Rules	All ingested feedback must be anonymized and pass PDPL and NCA compliance. Every amendment proposal must cite the supporting feedback excerpts. No amendment may be auto-applied — each requires explicit Analyst approval.
Data & Outputs	Inputs: Public consultation comments, inter-ministerial submissions. Outputs: Consultation Report, Thematic Clusters, Amendment Proposals, new Draft Version.
Acceptance Criteria	GIVEN a consultation dataset WHEN synthesis is triggered THEN themes are clustered with supporting excerpts AND amendment proposals are traceable to source feedback AND PDPL-compliant redaction is applied on ingestion.

Data Sections — UC-05



#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Consultation Feedback Synthesis						
Section: Intake						
1	Channel Filter	Multi-select	All	Optional	Absher / Email / Inter-Ministerial	—
2	Date Range	Date Range	Consultation window	Mandatory	Bounds ingestion period	—
3	PDPL Redaction Status	Badge	Enabled	—	Confirms auto-redaction active	Blocks ingestion if disabled
Section: Thematic Clusters						
4	Theme Table	Table	—	—	Theme, volume, sentiment, representative excerpts	—
5	Material-Threshold Flag	Badge	—	—	Shown if theme > 25% of feedback	Auto-escalation notification
Section: Amendment Proposals						
6	Proposal Row	Table	—	—	One per theme with linked clause + diff preview	—
7	Accept / Modify / Reject	Action (per row)	—	Mandatory	Analyst decision per proposal	Modify opens diff editor
8	Generate New Version	Button	—	Mandatory	Produces updated draft	Confirmation toast

UC ID	UC-06
Use Case Name	Executive Policy Review & Approval (HILT Gateway)
Related Functional Requirement	Refer To FR-14
Objective	Provide the Minister and Executive Office with a single decision-ready interface to review, approve, reject, or request revision of simulated policies, with full traceability.



Primary Actors	Minister (His Excellency)
Secondary Actors	Senior Leaders; Director of Policy; Human-in-the-Loop Governance System; Ministry of Finance (for Strategic Waivers)
Trigger / Precondition	A policy draft completes simulation with a Predictive Success Score $\geq 80\%$ and passes all Guardrails.
Preconditions	Draft has cleared Conflict Detection, Simulation, and Consultation stages; Executive Cockpit is accessible via Nafath; digital signature infrastructure is operational.
Main Scenario	- Minister opens the Executive Intelligence Cockpit and reviews the Policy Health dashboard. - Minister selects a policy and views the one-page summary produced by the Decision Justification Generator. - System presents Vision 2030 alignment score, fiscal projection, regional impact heat-map, top-three risks, and up to three simulated Recovery Scenarios if applicable. - Minister selects one of: Approve / Reject / Request Revision / Request Further Simulation. - Decision is logged immutably, digitally signed, and routed downstream (enactment, revision, or archival).
Alternative Scenarios	A1. If the Minister requests further simulation, the draft returns to the Housing Economist with new parameters. A2. If the Minister approves and a Strategic Waiver is required (e.g., negative NPV), the system routes the approval to the Ministry of Finance for co-signature. A3. If the system detects a conflict between the Minister's decision and an active Guardrail, it prompts an override justification before accepting the action. If failure occurs: the decision is held in a pending state and the Executive Office is notified to retry.
Business Rules	All ministerial decisions are digitally signed and immutable. Every decision must have an attached rationale stored in the Institutional Memory Vault. Override of an active Guardrail requires an explicit "Strategic Waiver" entry in the audit trail.
Data & Outputs	Inputs: Approved draft, simulation outputs, alignment score, fiscal & risk reports. Outputs: Decision Record, Digital Signature, Audit Log entry, downstream routing instruction.
Acceptance Criteria	GIVEN a policy that has cleared all prior gates WHEN the Minister renders a decision THEN the action is digitally signed AND logged immutably AND correctly routed (enactment / revision / archival) with full end-to-end traceability.

Data Sections — UC-06

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Executive Intelligence Cockpit						
Section: Header						
1	National Policy Health	Display (Gauge)	—	—	Composite national score	—
2	Vision 2030 Trajectory	Trend Chart	—	—	Vs. target	—



Section: Decision Queue						
3	Policy Tile	Tile	—	—	One tile per policy awaiting decision	Critical alerts highlighted
4	One-Page Summary	Expandable Panel	Collapsed	—	Decision Justification Generator output	—
5	Decision	Radio Group	—	Mandatory	Approve / Reject / Request Revision / Request Further Simulation	—
6	Digital Signature	Signature (Nafath)	—	Mandatory	Binds decision to Minister identity	—
Section: Recovery Scenarios (Conditional)						
7	Scenario Selector	Radio Group	—	Conditional	Shown when policy has simulated Recovery Scenarios	—
Section: Actions						
8	Request Waiver (MoF)	Button	—	Conditional	Enabled when Strategic Waiver required	Opens MoF co-sign workflow

UC ID	UC-07
Use Case Name	Regional Impact Visualization (Geospatial Heatmap)
Related Functional Requirement	Refer To FR-07
Objective	Provide Senior Leaders and regional directors with geospatial heatmaps showing how a policy's projected impact varies across Saudi Amanas and regions, enabling location-specific decision-making.
Primary Actors	Senior Leader / Executive
Secondary Actors	Policy Analyst; Housing Economist; Geospatial Rendering Engine (System); Balady (regional data feed)
Trigger / Precondition	User selects an active or simulated policy and clicks "View Regional Impact".
Preconditions	Policy has completed at least one simulation run; Balady regional data feed is current; user has role-based access to the affected regions.



Main Scenario	1. User selects the policy and the metric to visualize (homeownership, affordability index, fiscal load, supply velocity). 2. System queries the simulation output aggregated by Amana / region. 3. Geospatial Rendering Engine produces an interactive heatmap with colour-coded severity and drill-down tooltips. 4. User filters by urban / rural, income segment, or time horizon; system updates the heatmap in real time. 5. User exports the visualization or bookmarks the view into a Decision Deck.
Alternative Scenarios	A1. If a region has insufficient data (< minimum sample threshold), the heatmap displays a “Low Confidence” overlay for that Amana. A2. If a region is outside the user’s access scope, it is rendered as “Restricted” (grey) with no drill-down available. If failure occurs: the map falls back to a tabular regional summary and the geospatial service failure is logged.
Business Rules	Regional visibility is enforced by the user’s role and geographic access scope. All exported visualizations carry the simulation timestamp and confidence band watermark. Regional data more than 30 days old triggers a “Stale” badge (Data Latency Guard).
Data & Outputs	Inputs: Policy ID, metric selection, time horizon, filter selections; Balady regional dataset; simulation outputs. Outputs: Interactive heatmap, regional summary table, exportable PDF / PNG, audit log entry.
Acceptance Criteria	GIVEN a simulated policy WHEN the user opens the Regional Impact view THEN a heatmap renders within 5 seconds AND Amanas outside scope are marked Restricted AND low-confidence regions are flagged AND exports carry a confidence watermark.

Data Sections — UC-07

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Regional Impact Visualization						
Section: Header						
1	Policy Name	Text (Display)	—	—	Displays the selected policy name and version	—
2	Simulation Run ID	Text (Display)	—	—	Displays the simulation run reference	—
3	Last Updated	DateTime (Display)	—	—	Shows simulation run timestamp	Stale badge shown if > 30 days
Section: Filters						
4	Metric Selector	Dropdown	Homeownership	Mandatory	Options: Homeownership, Affordability, Fiscal, Supply Velocity	Updates heatmap on change



5	Time Horizon	Dropdown	12 months	Mandatory	Options: 6m / 12m / 24m / 60m	—
6	Urban / Rural Toggle	Toggle	All	Optional	Filters regions by urban classification	—
7	Income Segment	Multi-select	All	Optional	Filters by citizen income band	—
Section: Map						
8	Heatmap Canvas	Geospatial (Interactive)	—	—	Renders Amana-level impact, drill-down on click	Tooltip shows metric value + confidence
9	Legend	Display	—	—	Colour scale with min / max values	—
10	Confidence Overlay	Display (Conditional)	—	—	Grey hatch for restricted; amber for low-confidence	“Low Confidence” / “Restricted” tooltips
Section: Actions						
11	Export PDF	Button	—	Optional	Generates watermarked PDF with timestamp	Confirmation toast
12	Save to Decision Deck	Button	—	Optional	Bookmarks view to Executive Cockpit deck	Confirmation toast

UC ID	UC-08
Use Case Name	Role-Based Platform Access & Authentication
Related Functional Requirement	Refer To FR-08
Objective	Ensure every user (Minister, Analyst, Economist, Legal Officer, Data Scientist, Stakeholder) is authenticated via Nafath and granted feature access strictly based on their organizational role and RACI permissions.
Primary Actors	Any Authenticated User
Secondary Actors	Nafath SSO; MOMAH Enterprise Directory (AD / IAM); Audit Log Service
Trigger / Precondition	User opens the Policy AI Lab web portal.
Preconditions	User has a valid Nafath identity; user is provisioned in MOMAH AD; role mapping to capabilities is configured.



Data Sections — UC-08

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Login & Role-Based Landing						
Section: Nafath Login						
1	Nafath Trigger	Button	—	Mandatory	Initiates Nafath SSO redirect	—
2	MFA Code	Text (Masked)	—	Mandatory	OTP / biometric token from Nafath	Inline error on failure
3	Session ID	System (Hidden)	—	—	Issued on successful auth	—
Section: Role Selector (Conditional)						
4	Active Role	Dropdown	Primary role	Conditional	Shown only if user has multiple roles	“Select a role to continue”
Section: Session Status						



5	User Name	Text (Display)	—	—	Top-right badge, links to profile	—
6	Role Badge	Text (Display)	—	—	Displays active role + scope	—
7	Session Timer	Display	15:00	—	Counts down; warns at 14:00 idle	“Session expiring” toast at 14:00

UC ID	UC-09
Use Case Name	Black-Swan Stress Testing
Related Functional Requirement	Refer To FR-09
Objective	Enable Housing Economists and Data Scientists to stress-test a policy under extreme market conditions (interest rate shocks, material cost spikes, demographic disruptions) to measure resilience before enactment.
Primary Actors	Housing Economist
Secondary Actors	Data Scientist; Stress Test Engine (System); Policy Analyst (observer)
Trigger / Precondition	Economist selects an active or draft policy and launches a Black-Swan scenario.
Preconditions	Baseline simulation has been run successfully; stress scenario library is loaded; the economist has “Stress Test” permission.
Main Scenario	1. Economist selects baseline policy and chooses a stress scenario (from library or custom). 2. Economist configures shock parameters: interest rate delta, material cost delta, demographic shift magnitude. 3. System runs Monte Carlo over the stressed variable matrix. 4. System produces a Stress Test Report with resilience score, worst-case outcomes, and break-point thresholds. 5. Economist saves the report and optionally triggers a Tier-2 briefing if resilience is below tolerance.
Alternative Scenarios	A1. If the simulation fails convergence, the system auto-retries with relaxed Monte Carlo bounds and flags the result as “Non-Converged”. A2. If worst-case fiscal loss exceeds the Fiscal Hard-Stop, the system blocks enactment approval and notifies MoF liaison. If failure occurs: the run is aborted and diagnostic logs are forwarded to the Data Scientist.
Business Rules	Stress tests must preserve the baseline run so results are comparable. Any resilience score below 50 triggers mandatory Human-in-the-Loop review. Stress scenarios and their parameters are versioned for reproducibility.
Data & Outputs	Inputs: Baseline policy & simulation ID, stress scenario selection, shock parameters. Outputs: Stress Test Report, Resilience Score, Break-point Table, audit log entry.
Acceptance Criteria	GIVEN a baseline simulation WHEN a Black-Swan scenario is run THEN the Stress Test Report is produced AND the Resilience Score is displayed AND break-points below tolerance trigger HILT review.



#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Black-Swan Stress Test						
Section: Baseline						
1	Policy Reference	Lookup	—	Mandatory	Selects the baseline policy + simulation run	—
2	Baseline Predictive Score	Text (Display)	—	—	Pulled from last successful simulation	—
Section: Stress Scenario						
3	Scenario Template	Dropdown	Custom	Mandatory	Options include 2008-style shock, SIBOR spike, commodity shock	—
4	Interest Rate Delta (bps)	Number	0	Optional	Range -500 to +1000	Warns above +500 bps
5	Material Cost Delta (%)	Number	0	Optional	Range -50 to +200	Warns above +100%
6	Demographic Shock (%)	Number	0	Optional	Range -20 to +20	—
Section: Execution						
7	Run Stress Test	Button	—	Mandatory	Launches Monte Carlo run	Progress indicator + ETA
8	Resilience Score	Text (Display)	—	—	0–100, computed post-run	“Below tolerance” banner if < 50
9	Break-Point Table	Table (Display)	—	—	Lists variable thresholds where policy fails	—
Section: Actions						
10	Save Report	Button	—	Optional	Archives in Institutional Memory Vault	—



UC ID	UC-10
Use Case Name	Agent-Based Multi-Variable Simulation Configuration
Related Functional Requirement	Refer To FR-10
Objective	Let Data Scientists and Housing Economists configure the agent-based simulation lab with economic, social, and market agents, set their interaction rules, and launch integrated runs.
Primary Actors	Data Scientist
Secondary Actors	Housing Economist; Simulation Engine (System); Model Registry (System)
Trigger / Precondition	User opens the Simulation Lab and selects “Configure New Run”.
Preconditions	Model Registry contains at least one valid agent library; data feeds (SAMA, GASTAT, MoF) are current.
Main Scenario	1. User selects the policy mechanism to be modelled. 2. User configures active agent classes (Citizens, Developers, Banks, Municipalities) and populations. 3. User sets interaction rules and feedback loops via the rule editor. 4. User binds each agent variable to a data source (SIBOR, CPI, Demographics) from the registry. 5. User runs the simulation; system executes ABM, logs every agent step, and produces outputs.
Alternative Scenarios	A1. If any bound data source is stale, the simulation runs but the output is tagged “Low Confidence”. A2. If interaction rules contain logical contradictions, the system blocks run and highlights conflicts. If failure occurs: partial logs are preserved and the Data Scientist is alerted for model diagnostics.
Business Rules	Every configuration must be saved as a reproducible model version in the registry. Simulation can produce no more than 3 concurrent runs per user to manage compute load. Level-1 results must return in under 60 seconds (NFR-02).
Data & Outputs	Inputs: Policy mechanism, agent populations, interaction rules, variable bindings. Outputs: Simulation results (fiscal / social / market), Model Version ID, full agent log, performance metrics.
Acceptance Criteria	GIVEN a validated ABM configuration WHEN “Run” is triggered THEN Level-1 results return within 60 seconds AND the Model Version ID is saved AND all outputs are traceable to the configuration used.

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Simulation Lab — Configure Run						
Section: Policy Context						



1	Policy Mechanism	Lookup	—	Mandatory	Selects mechanism being modelled	—
2	Model Version Name	Text Input	v0.1	Mandatory	User-provided version label	Unique check
Section: Agent Configuration						
3	Citizens Population	Number	100000	Mandatory	Range 1,000–1,000,000	—
4	Developers Population	Number	500	Mandatory	Range 10–10,000	—
5	Banks Population	Number	20	Mandatory	Range 5–100	—
6	Municipalities Population	Number	13	Mandatory	Matches Saudi Amanas	—
Section: Rule Editor						
7	Interaction Rule Canvas	Visual Editor	—	Mandatory	Drag-and-drop rule composition	Conflict detection inline
8	Feedback Loop Toggle	Multi-select	Enabled	Optional	Enables macro feedback loops	—
Section: Variable Bindings						
9	Interest Rate Source	Dropdown	SAMA SIBOR	Mandatory	Bound to SAMA feed	Stale badge if > 30 days
10	Demographics Source	Dropdown	GASTAT	Mandatory	Bound to GASTAT	Stale badge if > 30 days
11	Inflation Source	Dropdown	GASTAT CPI	Mandatory	Bound to GASTAT CPI	—
Section: Execution						
12	Run Simulation	Button	—	Mandatory	Launches ABM execution	Progress bar + ETA
13	Concurrent Runs Indicator	Display	0 / 3	—	Blocks launch if at cap	“Queue full” on blocked launch

UC ID	UC-11
Use Case Name	Predictive Success Scoring & Failure Explanation



Related Functional Requirement	Refer To FR-11
Objective	Automatically score a simulated policy's probability of meeting its objectives (0–100%) and, where below the 80% threshold, produce a structured failure explanation with root causes and remediation hints.
Primary Actors	Policy Analyst
Secondary Actors	Housing Economist; Data Scientist; Predictive Scoring Engine (System)
Trigger / Precondition	A simulation run completes successfully.
Preconditions	Simulation outputs are available; historical outcome library contains at least 3 comparable precedents.
Main Scenario	1. Scoring Engine ingests simulation outputs and historical precedents. 2. System computes the Predictive Success Score based on KPI attainment probability. 3. If score $\geq 80\%$, policy advances to review; if $< 80\%$, Failure Explanation is generated. 4. Failure Explanation lists contributing factors ranked by impact and remediation hints. 5. Analyst reviews the score and explanation; may request re-simulation or escalate.
Alternative Scenarios	A1. If fewer than 3 precedents exist, the score is computed but labelled “Low Confidence” with widened bands. A2. If an outlier precedent dominates the score, the system flags it for Data Scientist review. If failure occurs: score computation fails gracefully, last-known score is displayed, and engineering is alerted.
Business Rules	Scores below 80% must include a structured Failure Explanation (XAI mandate). Scores and explanations are versioned and immutable. No policy may advance past review without a score.
Data & Outputs	Inputs: Simulation outputs, historical precedent library, KPI targets. Outputs: Predictive Success Score, Confidence Band, Failure Explanation (if applicable), audit entry.
Acceptance Criteria	GIVEN a completed simulation WHEN scoring runs THEN a Predictive Success Score is produced AND scores $< 80\%$ include a Failure Explanation AND all outputs are immutable and versioned.

Data Sections — UC-11

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Predictive Success Score						
Section: Score Panel						
1	Score Value	Display (Gauge)	—	—	0–100, colour-coded	—
2	Confidence Band	Display	—	—	Upper / lower bounds	“Low Confidence” badge if narrow precedent set



3	Status Badge	Display	—	—	Pass / Fail per 80% threshold	—
Section: Failure Explanation (Conditional)						
4	Contributing Factors	Table	—	Conditional	Ranked by impact weight	Displayed only when score < 80%
5	Remediation Hints	List	—	Conditional	Mechanism-level suggestions	—
Section: Precedents						
6	Precedent Table	Table	—	—	Lists matched historical policies	—
7	Outlier Flag	Display	—	—	Highlights if a precedent dominates	“Flagged for review” tooltip
Section: Actions						
8	Request Re-simulation	Button	—	Optional	Returns to Simulation Lab with prefilled params	—
9	Escalate to HILT	Button	—	Conditional	Enabled only for fail scores	Opens HILT ticket

UC ID	UC-12
Use Case Name	Risk Threshold & Guardrail Configuration
Related Functional Requirement	Refer To FR-12
Objective	Enable authorized leaders (Director, Chief Economist) to configure and maintain the risk thresholds, severity bands, and policy guardrails used by the automated Risk Escalation engine.
Primary Actors	Director of Policy / Chief Economist
Secondary Actors	Data Scientist; Audit Log Service; Risk Engine (System)
Trigger / Precondition	User opens the Risk Configuration panel.
Preconditions	User has “Risk Config” permission; current configuration is loaded from the registry.
Main Scenario	1. User opens the Risk Configuration panel and views the current threshold set. 2. User edits severity bands for one or more categories (Affordability Drift, Supply-Demand Gap, Fiscal Variance, Ownership Stagnation, Market Volatility). 3. User configures the escalation tier mapping for each category. 4. System validates the configuration against Guardrail logic and simulated historical



	data. 5. User saves; configuration is versioned and applied system-wide on next scan cycle.
Alternative Scenarios	A1. If a proposed threshold contradicts a Strategic Waiver, the system blocks save and prompts for MoF co-approval. A2. If the configuration disables a Critical-tier category, the system requires an explicit override justification. If failure occurs: changes are rolled back to the last valid version and the user is notified.
Business Rules	No Critical-tier category may be fully disabled without a logged override justification. All threshold changes are immutable and versioned in the audit trail. Changes apply to new scans only; in-flight evaluations complete with the prior configuration.
Data & Outputs	Inputs: Threshold values, severity bands, tier mappings, justifications. Outputs: New Configuration Version, Validation Report, audit log entry, change notification.
Acceptance Criteria	GIVEN a valid threshold change WHEN saved THEN the new configuration is versioned AND the next scan uses the new thresholds AND any override is logged with justification.

Data Sections — UC-12

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Risk Thresholds & Guardrails						
Section: Threshold Matrix						
1	Affordability Drift	Number	5.5	Mandatory	Price-to-income ratio trigger	Warns on increase > 20%
2	Supply-Demand Gap (%)	Number	15	Mandatory	Deficit threshold over 24 months	—
3	Fiscal Variance (%)	Number	10	Mandatory	Budget overrun without 1:1 ROI	—
4	Ownership Stagnation (%)	Number	2	Mandatory	Below Vision 2030 trendline	—
5	Market Volatility (%)	Number	20	Mandatory	Private developer investment drop YoY	—
Section: Tier Mapping						
6	Tier-1 Recipients	Multi-select	Policy Analysts	Mandatory	Medium severity	—



7	Tier-2 Recipients	Multi-select	Director / Chief Economist	Mandatory	High severity	—
8	Tier-3 Recipients	Multi-select	Minister / Exec Office	Mandatory	Critical severity	—
Section: Validation & Save						
9	Validate Against History	Button	—	Mandatory	Runs configuration against last 12 months of data	Shows alert count preview
10	Override Justification	Text Area	—	Conditional	Required if disabling a Critical category	Inline warning
11	Save Configuration	Button	—	Mandatory	Versions and applies the change	Confirmation toast

UC ID	UC-13
Use Case Name	Policy Lifecycle Orchestration (Draft → Enactment)
Related Functional Requirement	Refer To FR-13
Objective	Manage a policy's end-to-end lifecycle through enforced stage gates — Draft, Conflict Review, Simulation, Consultation, HILT Review, Executive Approval, Enactment, Post-Launch Monitoring — with full stage-history tracking.
Primary Actors	Product Manager / Business Owner
Secondary Actors	Policy Analyst; Legal Officer; Economist; Minister; Lifecycle Engine (System)
Trigger / Precondition	A policy is created in Draft status or advances from one stage to the next.
Preconditions	Lifecycle stages and entry/exit criteria are configured; roles are mapped to stage gates.
Main Scenario	1. Policy is created and enters "Draft" stage with a Product Manager as owner. 2. Product Manager moves the policy forward; system validates that exit criteria for the current stage are met. 3. Stage-appropriate approvers are notified; approvers review and approve / reject / request changes. 4. On approval, the policy advances to the next stage and the Stage History log is updated. 5. At "Enacted", the policy enters Post-Launch Monitoring where actual outcomes are tracked vs. simulated.
Alternative Scenarios	- A1. If exit criteria are not met, the system blocks advancement and lists unmet criteria. - A2. If a stage is rejected, the policy returns to the prior stage with reviewer comments. - A3. If a stage SLA is breached (no action within N days), the system auto-escalates to



	the stage owner's manager. If failure occurs: stage transition is rolled back atomically; no partial states are persisted.
Business Rules	Stage transitions are atomic — either fully applied or rolled back. Every transition is logged with actor, timestamp, and rationale. No stage may be skipped without an explicit governance waiver.
Data & Outputs	Inputs: Policy ID, target stage, exit criteria evidence, reviewer decisions. Outputs: Updated stage, Stage History entry, notifications, audit log entry.
Acceptance Criteria	GIVEN a policy meeting exit criteria WHEN advancement is requested THEN the stage transitions atomically AND the correct approvers are notified AND the Stage History is updated AND rejections return the policy to the prior stage with comments.

Data Sections — UC-13

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Policy Lifecycle Pipeline						
Section: Policy Header						
1	Policy Name	Text (Display)	—	—	Name + current version	—
2	Current Stage	Badge	—	—	Colour-coded by stage	—
3	Owner	Text (Display)	—	—	Product Manager assigned	—
Section: Stage Pipeline						
4	Pipeline Visual	Stepper (Display)	—	—	8 stages shown horizontally	Click a stage for history
5	Stage Owner	Text (Display)	—	—	Current approver/owner	—
6	SLA Countdown	Display	—	—	Days remaining in current stage	Warns at 48h to breach
Section: Stage History						
7	History Table	Table	—	—	Lists every transition with actor + reason	—
Section: Actions						
8	Advance Stage	Button	—	Conditional	Enabled only if exit criteria met	Lists unmet criteria on click if blocked



Section: Review Header						
1	Policy Name	Text (Display)	—	—	Name + draft version	—
2	Validator Role	Text (Display)	—	—	Displays active role + scope	—
3	Validation Progress	Progress Bar	0%	—	Percent of clauses reviewed	—
Section: Clause Review						
4	Clause Text	Text (Display)	—	—	AI-proposed clause	—
5	Explainability Report	Expandable Panel	Collapsed	—	Cites data / precedents / law	—
6	Decision	Radio Group	—	Mandatory	Accept / Edit / Reject / Kill-switch	Kill-switch requires justification
7	Edit Text	Text Area	—	Conditional	Shown if Decision = Edit	—
8	Justification	Text Area	—	Conditional	Required for Reject / Kill-switch	Inline validation
Section: Signature & Submission						
9	Digital Signature	Signature (Nafath)	—	Mandatory	Binds decisions to validator identity	—
10	Submit Review	Button	—	Mandatory	Enabled only when all clauses have decisions	—

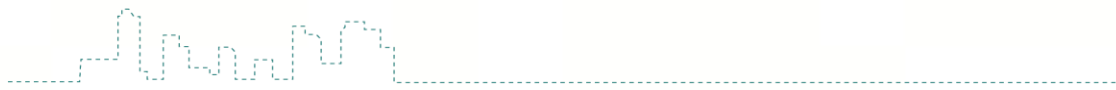
UC ID	UC-15
Use Case Name	Continuous Learning Loop — Model Retraining from Real-World Outcomes
Related Functional Requirement	Refer To FR-15
Objective	Close the loop between simulated and actual policy outcomes by continuously feeding real-world data back into the simulation models, improving predictive accuracy over time.
Primary Actors	Data Scientist



Secondary Actors	Housing Economist; Model Registry (System); Retraining Pipeline (System); Audit Log Service
Trigger / Precondition	Quarterly retraining schedule triggers, or Drift Detection flags a model as degraded.
Preconditions	At least 90 days of post-launch outcome data exist; model is not locked by an active simulation run.
Main Scenario	1. Retraining Pipeline gathers actual outcomes vs. simulated predictions for the evaluation window. 2. System computes drift metrics and accuracy deltas per model. 3. If drift exceeds tolerance, the pipeline retrains the model using updated training data. 4. System validates the retrained model against a held-out test set and regulatory ethics checks. 5. Data Scientist reviews the validation report and promotes the new model version to production, or rejects and rolls back.
Alternative Scenarios	A1. If the retrained model fails ethics / fairness checks, the promotion is blocked and a SDAIA-aligned review is initiated. A2. If data volume is insufficient, retraining is deferred and the model is tagged “Insufficient Data” until next cycle. A3. If the retrained model degrades accuracy, the pipeline auto-rolls back and preserves the incumbent. If failure occurs: pipeline state is preserved; partial artifacts are quarantined and the Data Scientist is alerted.
Business Rules	No model may be promoted to production without passing ethics / fairness checks (SDAIA). Every retraining run is reproducible: inputs, config, outputs are versioned. Rollbacks are one-click and preserve full lineage.
Data & Outputs	Inputs: Outcome data, simulated predictions, current model artifacts, training config. Outputs: Retrained Model Version, Validation Report, Drift Metrics, promotion/rollback decision, audit log entry.
Acceptance Criteria	GIVEN sufficient outcome data WHEN retraining runs THEN drift metrics are produced AND the retrained model is validated against ethics and accuracy checks AND promotion requires explicit Data Scientist approval AND rollback preserves full lineage.

Data Sections — UC-15

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Model Retraining & Promotion						
Section: Retraining Summary						
1	Model Name	Text (Display)	—	—	Model identifier + current version	—
2	Evaluation Window	Date Range	Last 90 days	Mandatory	Period of real-world outcomes used	—



3	Drift Score	Display	—	—	Composite drift metric 0–100	Warns above 70
Section: Validation						
4	Accuracy Delta	Display	—	—	New vs. incumbent accuracy	Colour-coded +/-
5	Ethics Check Status	Badge	—	—	Pass / Fail (SDAIA)	Blocks promotion on Fail
6	Test Set Results	Table	—	—	Per-scenario accuracy breakdown	—
Section: Promotion Decision						
7	Promote to Production	Button	—	Conditional	Enabled only if ethics pass and accuracy $\geq$ incumbent	Confirmation dialog
8	Rollback	Button	—	Optional	Restores incumbent immediately	Confirmation dialog
9	Promotion Rationale	Text Area	—	Mandatory	Required to log promotion reason	—

UC ID	UC-16
Use Case Name	Policy Gap & Opportunity Analysis
Related Functional Requirement	Refer To FR-04
Objective	Enable Policy Analysts and Strategists to automatically identify regulatory “white spaces” and under-served sectors where new policy interventions are required to meet Vision 2030 targets, and surface AI-generated opportunity recommendations grounded in KPI gaps and peer-country benchmarks.
Primary Actors	Policy Analyst / Strategist
Secondary Actors	Senior Leader; Housing Economist; Gap Analysis Engine (System); Vision 2030 PMO (data feed); Knowledge Graph (System)
Trigger / Precondition	User initiates a Gap Analysis run for a selected sector, sub-sector, or Vision 2030 pillar.
Preconditions	User is authenticated via Nafath with Strategist scope; Policy Registry is current; Vision 2030 KPI baseline and trendlines are loaded; global benchmark datasets (OECD, IMF, World Bank) are synced within the last 30 days.



Main Scenario	1. User selects the analysis scope (target sector, sub-sector, Vision 2030 pillar, region, time horizon). 2. System queries the Knowledge Graph for all active policies within the scope and maps their KPI coverage. 3. Gap Analysis Engine compares current coverage against Vision 2030 KPI targets and peer-country benchmarks. 4. System produces a Gap Report listing under-served areas, misaligned KPIs, and a 0–100 Opportunity Score per gap. 5. For each gap, the system proposes candidate policy interventions grounded in global benchmarks and linked to relevant legal basis. 6. User exports the report or sends a selected opportunity to the Drafting Workspace to initiate a new policy (UC-01).
Alternative Scenarios	A1. If benchmark datasets are older than 30 days, outputs are watermarked “Low Confidence” (Data Latency Guard). A2. If a proposed intervention conflicts with an existing policy, the system cross-links to the Regulatory Conflict Engine (UC-02) prior to drafting. A3. If a Critical gap is detected (e.g., KPI trajectory falling > 10% below Vision 2030 target), the system auto-escalates to a Tier-2 Strategic Briefing. If failure occurs: the analysis run is aborted, partial results are discarded, and the Data Scientist is alerted via a Drift Detection signal.
Business Rules	Every identified gap must cite at least one Vision 2030 KPI and at least one benchmark precedent (XAI mandate). Opportunity Scores are computed on a standardized 0–100 scale and are immutable per analysis run. Gaps flagged “Critical” cannot be dismissed without a logged justification stored in the audit trail.
Data & Outputs	Inputs: Scope selection, Vision 2030 KPI targets and trendlines, active Policy Registry subset, peer-country benchmark datasets. Outputs: Gap Report, Opportunity Scores, Benchmark Map, recommended interventions (with legal basis), audit log entry.
Acceptance Criteria	GIVEN a valid analysis scope WHEN Gap Analysis is triggered THEN the system identifies under-served areas within 60 seconds AND every gap is traceable to at least one Vision 2030 KPI AND at least one candidate intervention is proposed per Critical gap AND the report is stored with a confidence band.

Data Sections — UC-16

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Policy Gap & Opportunity Analysis						
Section: Scope Selection						
1	Target Sector	Dropdown	—	Mandatory	Options: Housing, Municipal, Infrastructure, Environmental, Social, Smart Cities	—



2	Vision 2030 Pillar	Multi-select	—	Mandatory	Maps to national Vision 2030 pillars	—
3	Time Horizon	Dropdown	24 months	Mandatory	Options: 12m / 24m / 60m	—
Section: Benchmark Settings						
4	Peer-Country Benchmarks	Multi-select	All	Optional	Options include Singapore, Norway, UAE, etc.	—
5	Benchmark Data Freshness	Badge	—	—	Displays age of peer-country datasets	“Low Confidence” badge if > 30 days
6	Run Gap Analysis	Button	—	Mandatory	Launches the Gap Analysis Engine	Spinner + ETA
Section: Gap Report & Actions						
7	Gap Row	Table	—	—	Each row = one under-served area with severity, Vision 2030 KPI linkage, and Opportunity Score	Critical rows highlighted
8	Proposed Intervention	Text (Display)	—	—	AI-generated candidate intervention per gap with legal basis	—
9	Send to Drafting Workspace	Button	—	Optional	Initiates a new policy draft from the selected gap	Opens UC-01 workspace

UC ID	UC-17
Use Case Name	Vision 2030 Alignment Auditing
Related Functional Requirement	Refer To FR-05
Objective	Provide Senior Leaders and Policy Analysts with a real-time quantitative score (0–100) measuring how strongly a policy draft contributes to Vision 2030 KPIs and



Data Sections — UC-17

#	Field Name	Field Type	Default Value	Mandatory	Business Rules	Messages / Notifications
Screen Title: Vision 2030 Alignment Audit						
Section: Draft Header						

1	Policy Reference	Lookup	—	Mandatory	Selects draft policy and version for auditing	—
2	Draft Version	Text (Display)	—	—	Current draft version number	—
3	Last Audit Timestamp	DateTime (Display)	—	—	Time of last scoring run	“Stale Baseline” badge if > 30 days
Section: Score Panel						
4	Composite Alignment Score	Display (Gauge)	—	—	0–100, colour-coded against threshold	“Below Threshold” banner if < 60
5	Pillar Breakdown	Bar Chart	—	—	Score per Vision 2030 pillar	—
6	Contribution Waterfall	Chart	—	—	Clause-by-clause contribution to the composite score	—
Section: Clause Drill-Down & Actions						
7	Clause Row	Table	—	—	Each row = one clause with its score and linked Vision 2030 KPI	Net-negative rows highlighted
8	Improvement Hint	Text (Display)	—	—	AI-suggested rewrite for low-scoring clauses	—
9	Return to Drafting	Button	—	Conditional	Enabled only if composite score < 70	Opens UC-01 Drafting Workspace

14.3 Screens (Optional):

None Available



15. Assumptions, Constraints, Prerequisites and Dependencies

15.1 Assumptions

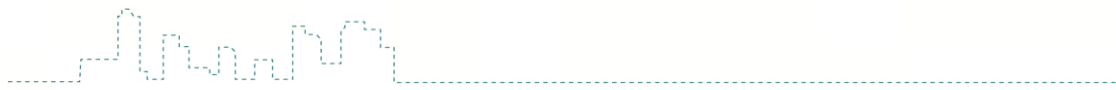
- Historical MOMAH policy data, decisions, and outcomes for the past five years are accessible and can be ingested into the Institutional Memory Vault during Phase 1.
- Integrated datasets from GASTAT, SAMA, Ministry of Finance, Balady, Sakani, and REGA will be available through formal data-sharing agreements and refreshed at least every 30 days.
- Technical and product teams (MOMAH, partner vendors, and SDAIA platform engineers) are allocated full-time for the duration of Phases 1–3.
- Subject-matter experts from Policy, Legal, Economics, and Municipal Affairs units will be available for requirements validation, model review, and UAT.
- Required SDAIA-approved sovereign AI infrastructure has sufficient capacity to host the simulation engine, knowledge graph, and inference workloads within the Kingdom.
- Executive sponsorship from His Excellency the Minister is sustained throughout Phase 1 roll-out to ensure change management and cross-ministerial cooperation.
- End users have access to devices compatible with the target platform — modern web browsers for analysts, and mobile & tablet devices for regional directors and executives.

15.2 Constraints

Type	Description
Technical	Must run on SDAIA-approved sovereign infrastructure within the Kingdom; integrate with existing MOMAH enterprise authentication (Nafath); compatibility required with Balady, Sakani, and REGA APIs; dependence on batch feeds from external ministries (real-time not always available); Arabic and English bilingual UI required across all modules.
Budget	Capped within the approved MOMAH FY26 digital-transformation envelope; any scope expansion requires re-approval by the Digital Transformation Steering Committee and, for cross-ministerial scope, co-approval by the Ministry of Finance.
Regulatory	Full compliance with Saudi PDPL (Personal Data Protection Law), NCA Essential Cybersecurity Controls (ECC), SDAIA AI Ethics Guidelines, and sovereign data-residency mandates. All AI inference and data storage must remain within the Kingdom; all outputs must pass the automated PDPL / NCA Compliance Gateway before release.
Timeline	Phase 1 (Registry & Drafting) delivery is tied to the FY26 Vision 2030 mid-cycle review; phased go-live must align with MOMAH quarterly governance cadences; executive-facing modules must be available before the next annual National Transformation Program reporting cycle.
Human Resources	Specialised skills in Agent-Based Modelling, Explainable AI, and Arabic NLP are scarce domestically; delivery depends on a blended team of MOMAH staff, SDAIA resources, and accredited external vendors, with knowledge-transfer obligations built into each phase.

15.3 Prerequisites

- Approved Data-Sharing Agreements (DSAs) with GASTAT, SAMA, Ministry of Finance, Balady, Sakani, and REGA are signed and operational.



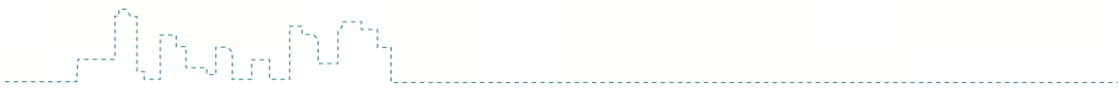
- Sovereign cloud environment provisioned and accredited by SDAIA and NCA.
- Nafath SSO integration with MOMAH identity provider is configured and tested.
- Canonical Policy Registry schema (Artifact 1.0) is validated and signed-off by Legal, Policy, and Data Governance functions.
- Stakeholder RACI and permissions matrix approved by the Policy AI Lab Steering Committee.
- Baseline Vision 2030 KPI dataset loaded for alignment scoring.
- Training and change-management plan for Policy Analysts, Economists, and Executives approved and resourced.

15.4 Dependencies

- SDAIA AI Platform availability and service-level agreement for sovereign model hosting and inference.
- Yesser Government Service Bus for inter-ministerial data exchange (MoF, SAMA, MISA).
- GASTAT Open Data API stability and update cadence.
- SAMA mortgage / interest-rate feed and Ministry of Finance macroeconomic forecast feed.
- Balady (municipal) and Sakani (housing) platform APIs, and the reliability of their data-refresh schedules.
- REGA real-estate regulatory database access.
- Completeness of the Legal Repository — timely ingestion of Royal Decrees, Cabinet resolutions, and ministerial circulars.
- Executive Office endorsement of risk thresholds, escalation tiers, and Guardrail parameters.
- Vision 2030 Program Management Office KPI definitions and annual trendline baselines.
- Availability of CSPO / Scrum-certified delivery leads and Arabic-NLP engineers for Phase-2 and Phase-3 modules.

16.0 Risks of Project

Risk Classification	[] High Risk [] Limited Risk [] Minimal Risk
Inaccurate or biased model outputs, difficulty integrating diverse data sources	High – Impacts all citizens of KSA
Potential misuse of generated scenarios, low adoption among policy teams	High – No trust in solution will lead to no adoption



Appendix

Appendix: RACI Matrix & Permissions

High-Level Permissions Summary

- [Permission summary point 1]
- [Permission summary point 2]

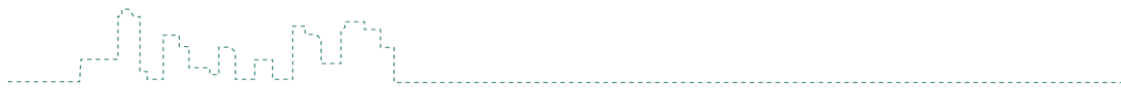
RACI Matrix — Key Processes

Process	User 1	User 2	User 3	User 4

Condensed Permissions Matrix

Action	User 1	User 2	User 3	User 4	User 5	User 6	User 7
	✓	View	—	—	—	View	View
	✓	✓	—	—	—	View	View
	✓	✓	—	—	—	View	View
	✓	View / Select	—	—	—	View	View
	✓	✓	—	—	—	View	View
	✓	✓	—	—	—	View	View
	View	✓	—	—	—	View	View
	✓	✓	—	—	—	View	View
	✓	✓	—	—	—	View	View





Approvals

Version	Name / Role	Approval Date	Signature